Anchoring Cliques in Clique Wrappers

A supplement to "A Least Squares Framework for the Maximum Weight Clique Problem"

Research notes prepared for Stanislav Busygin
with Claude (Anthropic)

September 26, 2026; revised October 18, 2026

Abstract

The non-edge entries of a wrapping matrix are free parameters of the clique wrapper. *Anchoring* spends them on a chosen maximal clique Q so that Q satisfies the hypothesis of [CW, Theorem 2]; its indicator then becomes a stationary point of the relaxed least squares program and lies on the curve that the secular equation traces. We collect what was established about anchoring, theoretically and in experiments with QUALEX-MS. For a single clique we give the construction, show that the indicator is then a point of the secular curve, and compute the curvature of the wrapper towards every other clique, which explains why anchoring acts as a local exchange around Q; the experiments confirm this and show that anchoring pays off as a second pass on a good clique and not otherwise. For several cliques the levels at which they can be anchored are forced: two anchored cliques Q_a, Q_b satisfy $\mu_a W(Q_b \setminus Q_a) = \mu_b W(Q_a \setminus Q_b)$, so cliques of equal weight share one level, and a shared level creates exactly the degenerate case (multiple eigenvalues with vanishing linear form) that the secular equation cannot see. We characterize when two cliques can be anchored together. Over all wrappers the best value of the relaxed program is $1/\vartheta(\bar{G}, w)$, the reciprocal of the Lovász number, and the standard wrapper gives Hoffman's bound on regular graphs; a clique indicator can be a global minimizer of the relaxed program only if $\omega(G, w) = \vartheta(\bar{G}, w)$, in which case every Lovász-optimal wrapper anchors all maximum cliques at the top of the spectrum, in the hard case of the trust region problem. Odd holes and odd antiholes admit no nontrivial anchoring of all their maximum cliques, which with the strong perfect graph theorem characterizes perfect graphs. For the quasigroup completion graphs of Busygin and Pasechnik the standard wrapper is itself Lovász-optimal on every completable instance, and the completions are exactly the nonnegative points of its degenerate sphere of global minimizers; this is why the hard case cannot be exploited. Busygin's SAT01, which recasts many NP problems as recognizing $\alpha < \vartheta = m$ in a weighted graph, generalizes this: an equation wrapper is Lovász-optimal exactly when $\vartheta = m$, which holds exactly when the instance has a vector solution, and it is blind to the contradictions outside the equations. For a proper Lovász-optimal wrapper the nonnegative points of its sphere of global minimizers are exactly the maximum clique indicators, the vertices of largest norm of a polytope; the hardness of SAT01 lies in finding them and not in a lack of properness, since for exact cover the equation wrapper is proper and the nonnegative points of its sphere are the exact covers. Every solution also lies on the surface of the standard wrapper, which meets that sphere and the orthant exactly in the solutions; the intersection of the sphere with a wrapper surface is its intersection with a cone, explicit in two unit vectors, so that of these conditions only nonnegativity is hard. Shrinking the degenerate eigenspace does not help either: since H_A is

Lovász-optimal, the Lovász optima form a spectrahedron over its top eigenspace, exact sparse elimination and a small interior point method give the smallest subspace that holds them all, with a certificate of minimality, and on all but the smallest instances that subspace keeps 91 to 95% of the eigenspace. Computations with a maximal clique enumerator and with CSDP illustrate the results on the DIMACS graphs, on random graphs, on random quasigroups with holes, and on SAT01 instances of factoring and of the Hamilton cycle problem, where ϑ refutes the smallest primes that propagation leaves open, with a vanishing margin, and none of the non-Hamiltonian graphs.

Contents

1 Introduction **3**

2 Setting **5**
2.1 Wrapping matrices . . . 5
2.2 The relaxed program and its stationary points . . . 6
2.3 The gauge . . . 7

3 Anchoring one clique **7**

4 Anchoring one clique in QUALEX-MS: experiments **9**
4.1 Protocol and provenance . . . 9
4.2 The construction and the spectrum . . . 10
4.3 Cold, warm and iterated anchoring . . . 10
4.4 Blending in one step of $\lambda_{\max}$ minimization . . . 10

5 Anchoring several cliques **11**

6 Anchoring every maximum clique **14**
6.1 The best relaxation is the Lovász number . . . 14
6.2 Global anchoring . . . 15
6.3 Colourings, holes and perfect graphs . . . 16
6.4 Quasigroup completion: why the hard case does not help . . . 17
6.5 SAT01: exact covers with contradictions . . . 19
6.6 Where the hardness lies . . . 21
6.7 A second wrapper surface . . . 23
6.8 The smallest jam . . . 26

7 Computations **28**
7.1 Tools . . . 28
7.2 Anchoring all maximum cliques of DIMACS graphs . . . 28
7.3 Random graphs . . . 30
7.4 The Lovász number of the 14 undecided graphs . . . 30
7.5 Relaxed bounds of the standard and the optimal wrapper . . . 30
7.6 Quasigroups with holes . . . 30
7.7 SAT01 instances . . . 32

8 Conclusions and open questions **36**

1 Introduction

This note supplements [CW], whose notation it follows, and refers to its statements as [CW, Theorem 2] and so on. The same notation is used in QUALEX-MS [Bu06], whose trust region stage finds the stationary points of [CW, Section 5]; Remark 2.4 gives the dictionary.

[CW, Theorem 2] says that a maximal clique Q is recognized through a stationary point of the relaxed least squares program [CW, (7)] whenever every vertex outside Q is attached to Q with the same weight, measured through the wrapping matrix H [CW, (6)]:

$$\sum_{j \in Q} h_{ij} z_j = C z_i \quad \forall i \in V \setminus Q. \tag{1}$$

For the standard wrapping matrix this is a strong combinatorial condition ([CW, Corollary 1]), and it rarely holds. But the entries of H on non-adjacent pairs are free, and (1) is linear in them. *Anchoring* Q means choosing them so that (1) holds. (The solver's sources and diagnostics call (1) "Theorem 8".)

The questions studied were:

1. What does anchoring one clique do to the stationary points, and is it useful in QUALEX-MS? (Sections 3 and 4.)

2. Can several cliques, in particular all maximum weight cliques, be anchored in one wrapper, and what does such a wrapper look like? (Sections 5 and 6.) For perfect graphs one expects the Lovász-optimal wrapper to anchor all maximum cliques at $\mu = \lambda_{\max}$; for general graphs the conditions look overdetermined.

3. How do these statements look on actual graphs? (Section 7.)

4. What do they say about SAT01 [Bu07], where deciding $\alpha = \vartheta$ in a weighted graph captures many NP problems, where does its hardness lie once ϑ and a Lovász-optimal wrapper are known, and how much does shrinking that wrapper's degenerate eigenspace help? (Sections 6.5, 6.6, 6.7, 6.8 and 7.7.)

The main findings, in the order of the text:

- Anchoring one maximal clique is always possible, lowering entries only (Proposition 3.4); it makes the indicator of Q the point of the secular curve at the multiplier $\mu_Q = W(Q) - C$ (Lemma 3.3). The curvature of the wrapper along the direction from x^Q to any other clique Q' exceeds μ_Q by exactly $\mu_Q(W(Q') - W(Q))/W(Q \triangle Q')$, whatever the free entries are (Proposition 3.5).

- In QUALEX-MS, anchoring the clique found by a standard pass and solving again never lost on 366 test graphs and improved 55 of 120 weighted random graphs; anchoring the greedy clique lost far more often than it won (Section 4).

- Two cliques anchored in one wrapper satisfy the *level identity* $\mu_a W(Q_b \setminus Q_a) = \mu_b W(Q_a \setminus Q_b)$ (Theorem 5.1). Cliques of equal weight therefore share a level, and a shared level forces an eigenvalue of $\hat{H}$ whose eigenspace contains the differences of their indicators and carries no linear form (Theorem 5.3): the degenerate case excluded in [CW, Section 5]. Theorem 5.4 characterizes when two cliques can be anchored together; two maximum

weight cliques always can (Corollary 5.5), though some non-edges may have to be made invisible (Proposition 5.6).

- $\max_H \min\{x^T x : x \in \mathcal{W}^H\} = 1/\vartheta(\bar{G}, w)$ (Theorem 6.1); for regular graphs the standard wrapper attains Hoffman's ratio bound instead (Corollary 6.2). A clique indicator is a global minimizer of the relaxed program for some wrapper iff $W(Q) = \omega(G, w) = \vartheta(\bar{G}, w)$ (Theorem 6.3); then every Lovász-optimal wrapper anchors every maximum clique at $\mu = \lambda_{\max}(\hat{H})$, the hard case of the trust region problem (Corollary 6.4).

- Anchoring all maximum cliques of an odd hole or an odd antihole forces the level to 0 (Theorem 6.6), while graphs with $\chi = \omega$ have an explicit Lovász-optimal wrapper anchoring them all (Proposition 6.5). Hence a graph is perfect iff every induced subgraph admits an anchoring of all its maximum cliques at a nonzero level (Theorem 6.7).

- Of the 31 DIMACS graphs with $n \leq 1100$ and between 2 and 20000 maximum cliques, 11 do not admit anchoring all of them at a nonzero level; of the remaining 20, 14 have $\vartheta = \omega$ and 6 have $\vartheta > \omega$ (Section 7).

- For the quasigroup completion graphs of [BP03] the standard wrapper is Lovász-optimal on every completable instance, and the completions are exactly the nonnegative points of its sphere of global minimizers (Proposition 6.10); finding one is QCP. On random quasigroups with holes its degenerate eigenspace was up to 55 times the span of the completions (Section 7.6).

- For a SAT01 instance with m equations the equation wrapper H_A generalizes that wrapper: it is Lovász-optimal iff $\vartheta = m$, then anchors every solution at the top, and its nonnegative global minimizers are the exact covers of the equations, whatever the other contradictions (Proposition 6.11). $\vartheta = m$ iff the instance has a vector solution, and the Lovász optima are the Gram matrices of the vector solutions (Theorem 6.12). After SAT01 propagation ϑ refutes four of the six primes below 2^{16} that propagation leaves open, with a margin falling from 0.064 to $7 \cdot 10^{-5}$, none of them without the contradictions that propagation derives, and none of the non-Hamiltonian graphs that survive it. Up to 15 bits the Lovász optimum of a satisfiable factoring instance is spanned by its solutions, a unique solution being the only optimum, while the two solutions of $241 \cdot 251$ sit in an optimal face of dimension 214 (Sections 6.8 and 7.7).

- For a proper Lovász-optimal wrapper the nonnegative points of the sphere of global minimizers are exactly the maximum clique indicators, the vertices of largest norm of the polytope of nonnegative points of the top eigenspace, and they exist iff $\omega = \vartheta$ (Proposition 6.13). This does not make SAT01 easy: the nonnegative points of the equation wrapper's sphere are the exact covers, so deciding whether such a sphere has one is NP-complete, although H_A is proper when every contradiction lies in an equation and Lovász-optimal on every satisfiable instance (Corollary 6.14). With contradictions outside the equations a nonnegative non-clique point of the optimal face makes every Lovász-optimal wrapper improper (Proposition 6.15), and the Hamilton cycle instances have such points (Section 6.6).

- Every solution also lies on the surface $\mathcal{W}^{H_0}$ of the standard wrapper, and the sphere of H_A, the orthant and $\mathcal{W}^{H_0}$ meet exactly in the solutions (Proposition 6.16). The intersection of

a sphere with any wrapper surface is its intersection with a cone, given in closed form by two unit vectors (Proposition 6.18). Added to QUALEX-MS's Douglas–Rachford stage, the surface solves one more factoring instance with the orthant and adds nothing to the projection that respects the contradictions outside the equations; without nonnegativity it finds nothing at all (Section 6.7).

- When $\vartheta = m$ the Lovász optima are the trace-one positive semidefinite matrices of a linear space V over the top eigenspace L_A of H_A. A certificate that the subspace R^* spanned by them all is the smallest is the first-order part of a perturbation of H_A in its free entries (Proposition 6.19). Exact sparse elimination gives V, and an interior point method in its independent constraints gives R^* with such a certificate, also on $1000001 = 101 \cdot 9901$, which is beyond CSDP. R^* is the span of the solutions on 14471, $179 \cdot 181$ and the cubic graph with 16 vertices, but keeps 91 to 95% of L_A on $241 \cdot 251$, on 1000001 and on the other Hamilton cycle instances, and the smaller sphere does not help the Douglas–Rachford search (Section 6.8).

2 Setting

2.1 Wrapping matrices

Let $G(V, E)$, $V = \{1, \dots, n\}$, have positive vertex weights w, let $z_i = \sqrt{w_i}$ and $W(S) = \sum_{i \in S} w_i$, and let $N(i)$ be the neighborhood of i. For $S \subseteq V$ we write $\mathbf{1}_S$ for its characteristic vector, z_S for the vector with $(z_S)_i = z_i$ on S and 0 elsewhere, and $x^S = z_S / W(S)$ for its indicator [CW, Definition 1].

Definition 2.1. $\mathcal{H}(G, w)$ is the set of real symmetric $n \times n$ matrices H with $h_{ij} = z_i z_j$ whenever $i = j$ or $(i, j) \in E$. For $H \in \mathcal{H}(G, w)$ and a non-adjacent pair $i \neq j$ we write

$$h_{ij} = z_i z_j (1 - \tau_{ij}),$$

and call $\tau_{ij} = \tau_{ji}$ the *defect* of the non-edge. The standard wrapping matrix H_0 of [CW] has $\tau \equiv 1$. The wrapping matrices of [CW], which need $h_{ij} < z_i z_j$ off E, are the H with all defects positive; we call them *proper*.

Every $H \in \mathcal{H}(G, w)$ defines the surface $\mathcal{W}^H = \{x \in \mathbb{R}^n : x^T H x = 1,\ z^T x = 1\}$. The computation in the proof of [CW, Proposition 2] involves only entries of H inside the clique, so *every clique indicator lies on* $\mathcal{W}^H$ *for every* $H \in \mathcal{H}(G, w)$. Nonnegative defects are exactly what [CW, Proposition 1] needs.

The weighted Lovász number of the complement of G is

$$\vartheta(\bar{G}, w) = \min_{H \in \mathcal{H}(G,w)} \lambda_{\max}(H) \tag{2}$$

$$= \max\{\langle z z^T, X \rangle :\ X \succeq 0,\ \operatorname{tr} X = 1,\ X_{ij} = 0 \text{ for } i \neq j,\ (i, j) \notin E\}, \tag{3}$$

where both optima are attained [Lo79, GLS88, Kn94]. We call a matrix attaining (2) a *Lovász-optimal* wrapping matrix and a matrix attaining (3) a *Lovász optimum*. Since the two programs have equal values, every Lovász-optimal H and every Lovász optimum X satisfy the complementarity condition

$$(\vartheta(\bar{G}, w)\, I - H)\, X = 0, \tag{4}$$

because $\operatorname{tr}((\vartheta(\bar{G},w)I - H)X) = \vartheta(\bar{G},w) - \langle zz^T, X\rangle = 0$ (H and zz^T agree on the support of X) and both factors are positive semidefinite.

Lemma 2.2. *For every $H \in \mathcal{H}(G,w)$ and every clique Q, $z_Q^T H z_Q = W(Q)^2$. Hence $\lambda_{\max}(H) \geq W(Q)$ and $\omega(G,w) \leq \vartheta(\bar{G},w)$; and $X_Q = z_Q z_Q^T / W(Q)$ is feasible in (3) with value $W(Q)$.*

Proof. $z_Q^T H z_Q = \sum_{i,j\in Q} z_i z_j h_{ij} = \sum_{i,j\in Q} w_i w_j = W(Q)^2$ and $\|z_Q\|^2 = W(Q)$, so the Rayleigh quotient of z_Q is $W(Q)$. X_Q is positive semidefinite with trace 1 and is supported on $Q \times Q$, where all pairs are adjacent; its value is $(z^T z_Q)^2/W(Q) = W(Q)$. □

2.2 The relaxed program and its stationary points

For $H \in \mathcal{H}(G,w)$ let

$$v(H) = \min\{x^T x : \ x \in \mathcal{W}^H\}, \tag{5}$$

the relaxed program [CW, (7)]. It is feasible (clique indicators) and its objective is coercive, so the minimum is attained, and $v(H) \leq 1/\omega(G,w)$.

As in [CW, Section 5] let $x^0 = z/W(V)$, $P = I - zz^T/W(V)$, $\hat{H} = PHP$, $\hat{b} = PHx^0$ and $s = 1 - (x^0)^T H x^0$. Writing $x = x^0 + \hat{x}$ with $\hat{x} \perp z$,

$$x^T x = \|x^0\|^2 + \|\hat{x}\|^2, \qquad x^T H x = 1 \iff q(\hat{x}) := \hat{x}^T \hat{H} \hat{x} + 2\hat{b}^T \hat{x} = s.$$

In terms of the defects, $s = W(V)^{-2} \sum w_i w_j \tau_{ij}$, the sum running over ordered non-adjacent pairs $i \neq j$, so $s > 0$ for proper wrapping matrices. Throughout, eigenvalues of $\hat{H}$ are those of its restriction to $z^\perp$ (on z itself $\hat{H}$ vanishes).

With the multiplier convention of [CW] (the multiplier of the quadratic constraint written as $-1/\mu$), a point $x = x^0 + \hat{x} \in \mathcal{W}^H$ is *stationary with multiplier $\mu \neq 0$* if

$$(\mu I - \hat{H})\,\hat{x} = \hat{b} \quad \text{on } z^\perp. \tag{6}$$

In the eigenvector basis $\hat{H} = U \operatorname{diag}(\lambda_i) U^T$, $c = U^T \hat{b}$, $y = U^T \hat{x}$, (6) reads $(\mu - \lambda_i) y_i = c_i$. For μ not an eigenvalue, $x(\mu) = x^0 + (\mu I - \hat{H})^{-1}\hat{b}$ is the *secular curve*, and $x(\mu) \in \mathcal{W}^H$ iff μ solves the secular equation [CW, (19)].

Lemma 2.3. *Let $x = x^0 + \hat{x} \in \mathcal{W}^H$.*

(a) If x is stationary with a multiplier $\mu > 0$ such that $\mu \geq \lambda_{\max}(\hat{H})$, then x is a global minimizer of (5).

(b) If $s > 0$, $\lambda_{\max}(\hat{H}) > 0$ and x is a global minimizer of (5), then x is stationary with a multiplier $\mu \geq \lambda_{\max}(\hat{H})$.

Proof. (a) On $z^\perp$ the function $L(y) = \|y\|^2 - \mu^{-1}(q(y) - s)$ has the Hessian $2(I - \hat{H}/\mu) \succeq 0$, so it is convex, and $\nabla L(\hat{x}) = 0$ by (6). Hence for every feasible y (i.e. $q(y) = s$) $\|y\|^2 = L(y) \geq L(\hat{x}) = \|\hat{x}\|^2$.

(b) Let $r = \|\hat{x}\|$. If some $y \in z^\perp$ with $\|y\| < r$ had $q(y) \geq s$, then, since $q(0) = 0 < s$, $q(ty) = s$ for some $t \in (0,1]$, a feasible point of norm $t\|y\| < r$. So $q < s$ on the open ball of radius r and, by continuity, $\hat{x}$ maximizes q over the closed ball. By the characterization of trust region solutions [MS83] applied to $-q$ there is $\mu \geq 0$ with $(\mu I - \hat{H})\hat{x} = \hat{b}$ and $\mu I - \hat{H} \succeq 0$ on $z^\perp$; thus $\mu \geq \lambda_{\max}(\hat{H}) > 0$. □

Remark 2.4 (QUALEX-MS). The trust region program of [Bu06] (cf. [CW, Theorem 3]) maximizes x^TAx, $A = H - w_{\min}I$, over $z^Tx = 1$, $\|x\| = r$. On such a sphere x^Tx is constant, so its stationary points are those of q on $\|\hat{x}\| = \sqrt{r^2 - \|x^0\|^2}$; they satisfy (6) with the same $\hat{x}$. The solver works with $\hat{A} = PAP = \hat{H} - w_{\min}P$ and the multiplier $\mu^* = \mu - w_{\min}$: the same eigenvectors and linear form, all eigenvalues and multipliers shifted by $w_{\min}$. By Lemma 2.3, when $s > 0$ and $\lambda_{\max}(\hat{H}) > 0$ the global minimizers of (5) are the stationary points with $\mu \geq \lambda_{\max}(\hat{H})$, which are the global maximizers of the trust region objective on their own sphere [MS83]; in the solver's terms, $\mu^* \geq \lambda_{\max}(\hat{A})$. Counts of "eigenvalues above μ^*" reported by the solver are counts of eigenvalues of $\hat{H}$ above μ.

2.3 The gauge

Lemma 2.5. *For $H \in \mathcal{H}(G, w)$ and $t \neq -1$ let $H_t = (1+t)H - t\,zz^T$. Then $H_t \in \mathcal{H}(G, w)$ with defects $(1+t)\tau$, $\mathcal{W}^{H_t} = \mathcal{W}^H$ and $v(H_t) = v(H)$; moreover $\hat{H}_t = (1+t)\hat{H}$, $\hat{b}_t = (1+t)\hat{b}$, $s_t = (1+t)s$, and a point is stationary for H with multiplier μ iff it is stationary for H_t with multiplier $(1+t)\mu$.*

Proof. On E and the diagonal the entries of H_t are $(1+t)z_iz_j - tz_iz_j = z_iz_j$, and on a non-edge $(1+t)z_iz_j(1-\tau_{ij}) - tz_iz_j = z_iz_j(1-(1+t)\tau_{ij})$. For $z^Tx = 1$, $x^TH_tx = (1+t)x^THx - t$, which equals 1 iff $x^THx = 1$. Since $Pz = 0$, $\hat{H}_t = (1+t)\hat{H}$ and $\hat{b}_t = (1+t)PHx^0 - tPz(z^Tx^0) = (1+t)\hat{b}$, and $s_t = 1 - (1+t)(x^0)^THx^0 + t = (1+t)s$. Multiplying (6) by $1+t$ gives the last claim. □

So the wrapper surface depends only on the defects up to a common factor, and the multipliers are defined up to the same factor. In particular the uniform perturbation of the standard wrapper, $h_{ij} = -tz_iz_j$ on every non-edge, is $(H_0)_t$ and changes nothing. For $1 + t > 0$ the position of a multiplier within the spectrum of $\hat{H}$ and the ratio

$$\rho(H) = \frac{\lambda_{\max}(\hat{H}) - \lambda_{\min}(\hat{H})}{\|\hat{b}\|}$$

are invariant; absolute spectral widths are not.

3 Anchoring one clique

Definition 3.1. A maximal clique Q is *anchored at level μ* in $H \in \mathcal{H}(G, w)$ if (1) holds with $C = W(Q) - \mu$.

Lemma 3.2. *The following are equivalent:*

(a) Q is anchored at level μ in H;

(b) $\sum_{j \in Q \setminus N(i)} w_j\tau_{ij} = \mu$ for every $i \in V \setminus Q$;

(c) $Hz_Q = (W(Q) - \mu)\,z + \mu\,z_Q$.

Proof. For $i \notin Q$, $\sum_{j \in Q} h_{ij}z_j = z_i \sum_{j \in Q \cap N(i)} w_j + z_i \sum_{j \in Q \setminus N(i)} w_j(1 - \tau_{ij}) = z_i\big(W(Q) - \sum_{j \in Q \setminus N(i)} w_j\tau_{ij}\big)$, which gives (a) $\iff$ (b). For $i \in Q$, $(Hz_Q)_i = z_iW(Q)$, which is the i-th entry of the right-hand side of (c) for every μ; for $i \notin Q$ the i-th entry of (c) is (1). □

For the standard wrapper the sum in (b) is $W(Q \setminus N(i))$, the weight of the non-neighbours of i in Q, so Q is anchored in H_0 iff that weight is the same for all outside vertices, which is [CW, Corollary 1]. Note that a clique that is not maximal can only be "anchored" at level 0: a vertex adjacent to all of Q has an empty sum in (b).

Lemma 3.3. *Let Q be a maximal clique and $\mu \neq 0$. Then Q is anchored at level μ in H iff x^Q is a stationary point of (5) with multiplier μ. In that case $\hat{x}^Q = x^Q - x^0$ satisfies $(\mu I - \hat{H})\hat{x}^Q = \hat{b}$, and if μ is not an eigenvalue of $\hat{H}$ then $x^Q = x(\mu)$ is the point of the secular curve at μ and μ is a root of the secular equation.*

Proof. Since $z^T x^Q = 1$, $Px^Q = x^Q - x^0 = \hat{x}^Q$, and $PHx^Q = PH(x^0 + \hat{x}^Q) = \hat{b} + \hat{H}\hat{x}^Q$. If Q is anchored, Lemma 3.2(c) gives $Hx^Q = \frac{W(Q)-\mu}{W(Q)} z + \mu x^Q$, so $PHx^Q = \mu \hat{x}^Q$, which is (6). Conversely, (6) says $P(Hx^Q - \mu x^Q) = 0$, i.e. $Hz_Q = \mu z_Q + \kappa z$ for some κ; comparing an entry in Q gives $\kappa = W(Q) - \mu$, which is Lemma 3.2(c). □

So anchoring is exactly what makes the secular equation of [CW] find Q: the curve $x(\mu)$ passes through x^Q at $\mu = \mu_Q$. [CW, Theorem 2] is the direction "anchored ⇒ stationary"; for proper wrappers $\mu_Q > 0$ by Lemma 3.2(b).

Proposition 3.4 (the construction)**.** *Let Q be a maximal clique and $H \in \mathcal{H}(G, w)$. For $i \notin Q$ let $C_i = z_i^{-1} \sum_{j \in Q} h_{ij} z_j$ and $m_i = W(Q \setminus N(i)) > 0$. For any $C \leq \min_{i \notin Q} C_i$, the matrix H' obtained by*

$$h'_{ij} = h'_{ji} = h_{ij} + \frac{C - C_i}{m_i} z_i z_j \qquad (i \notin Q,\ j \in Q \setminus N(i)),$$

all other entries unchanged, lies in $\mathcal{H}(G, w)$, satisfies $H' \leq H$ entrywise, and anchors Q at level $W(Q) - C$. If H is proper so is H'.

Proof. A non-edge between $i \notin Q$ and $j \in Q$ enters condition (1) of i only, so the conditions can be met one outside vertex at a time; the correction raises C_i by $\frac{C-C_i}{m_i} \sum_{j \in Q \setminus N(i)} w_j = C - C_i$. The corrections are nonpositive, which lowers entries and raises defects. □

QUALEX-MS implements this as `anchor_wrapper()` with $C = \min_i C_i$, optionally damping all corrections by a factor $\theta \in (0, 1]$. For a single clique every level can be reached; $C \leq \min_i C_i$ is what keeps the corrections nonpositive.

Proposition 3.5 (curvature towards other cliques)**.** *Let Q be anchored at level μ in H, let $Q' \neq Q$ be any clique and $u = x^{Q'} - x^Q$. Then $u \perp z$ and*

$$\frac{u^T \hat{H} u}{u^T u} = \mu \left(1 + \frac{W(Q') - W(Q)}{W(Q \triangle Q')}\right).$$

Proof. Write $a = W(Q')$, $b = W(Q)$, $c = W(Q \cap Q')$. Since $z^T u = 0$, $u^T \hat{H} u = u^T H u$. By Lemma 2.2, $z_{Q'}^T H z_{Q'} = a^2$ and $z_Q^T H z_Q = b^2$; by Lemma 3.2(c), $z_{Q'}^T H z_Q = (b-\mu) z_{Q'}^T z + \mu z_{Q'}^T z_Q = (b - \mu)a + \mu c$. Hence

$$u^T H u = \frac{a^2}{a^2} + \frac{b^2}{b^2} - \frac{2((b-\mu)a + \mu c)}{ab} = \frac{2\mu(a-c)}{ab}, \qquad u^T u = \frac{1}{a} + \frac{1}{b} - \frac{2c}{ab} = \frac{a+b-2c}{ab},$$

and $a+b-2c = W(Q \triangle Q')$, so the quotient is $2\mu(a-c)/(a+b-2c) = \mu(1+(a-b)/(a+b-2c))$. □

Corollary 3.6. *Let Q be anchored at a level $\mu > 0$.*

(a) If some clique Q' is heavier than Q, then $\lambda_{\max}(\hat{H}) > \mu$: the anchored indicator is not a maximizer of the trust region program on its sphere, and the excess curvature towards Q' is the gain $W(Q') - W(Q)$ per unit of exchanged weight $W(Q \triangle Q')$, times μ.

(b) Towards a clique of the same weight the curvature equals μ; towards a lighter one it is smaller.

Part (a) is quantitative, not merely the observation that $x^{Q'}$ is a feasible point of smaller norm: the directions of positive excess curvature point to heavier cliques, most strongly to those that differ from Q by a small exchange with a gain. This is the local exchange behaviour measured in Section 4.

4 Anchoring one clique in QUALEX-MS: experiments

4.1 Protocol and provenance

A QUALEX-MS *pass* builds a wrapping matrix, runs the trust region stage and refines the stationary points into cliques. The variants (named as in `bench/variants`) were:

head the standard pass with H_0;

anc (*cold*) one pass with H_0 anchored on the clique of the greedy MIN heuristic; **anc05** the same with $\theta = 0.5$; **ancs** with the corrections dealt out to randomly permuted outside vertices, so that (1) fails (a control); **ancp** re-anchored on each improvement, at most 4 anchored passes;

ancw (*warm*) the standard pass, then one pass anchored on its final clique; **ancws** its shuffled control; **ancwp** re-anchored on each improvement, at most 4 anchored passes;

icel as ancw, then one line-searched step of $\lambda_{\max}(H)$ minimization along the gradient $2u_iu_j$ (u the top eigenvector) on all non-edges, entries clipped at $h_{ij} \leq 0$, and re-anchoring ("fire and ice"); **icels** with the step's entries shuffled over the non-edges; **ices** a step on $\lambda_{\max}(\hat{H})$ that only moves each outside vertex's correction among its non-neighbours in Q, so that (1) stays exact;

lz the standard pass replaced by a pass on the $\lambda_{\max}$-stepped standard wrapper; **lzw** that wrapper as a second pass after the standard one; **lzws** its shuffled control.

The test sets were the 70 DIMACS graphs outside the ten largest; a set of 56 weighted graphs; uniform random graphs $G(n, p)$ with $n \in \{200, 300, 500, 1000\}$, $p \in \{0.25, 0.5, 0.7, 0.8, 0.9, 0.95\}$, five of each, unweighted (`ru`) and with integer weights uniform on $\{1..10\}$ (`rw`), for 70 of the weighted and 50 of the unweighted of which cliquer [NÖ03] proved the optimum; and for the largest runs the ten largest DIMACS graphs and 24 random graphs with $n = 2000$.

Results quoted from the experiments of 12–13 September 2026 were measured before the benchmark tooling was put under version control; their logs, and the generator of the 56-graph weighted set, were lost when the machine's `/tmp` was wiped. Tables 1 and 2 are reproducible: `bench/compare.py` computes them from the results committed to the QUALEX-MS repository (directory `bench/runs`, solver built from commit 6cebb51), and they reproduce the corresponding earlier figures exactly.

4.2 The construction and the spectrum

The construction is exact: on every anchored run the stationary point at μ_Q returned the indicator to 10^{-15}. The anchored indicator was never the maximizer on its sphere: between 8 and 276 eigenvalues of $\hat{H}$ lay above μ_Q (median 30 of 399 on DIMACS). Taking C below $\min_i C_i$ does not change this: each unit taken off C raises μ_Q by one but $\lambda_{\max}$ by 1.2 to 2.6, and the count above μ_Q levels off at up to $|Q|$ (the correction is supported on the $Q \times (V \setminus Q)$ block, so it has at most $|Q|$ positive eigenvalues). By Theorem 6.3 below, no choice could have made the indicator a global maximizer on these graphs unless $\vartheta = \omega$.

The corrections are large (smallest $t_i = (C - C_i)/m_i$: median -7 on DIMACS, -10 on weighted graphs, down to -132), and forced: a vertex missing a single vertex of Q must put its whole correction on one entry. One anchoring widens the absolute spectrum of $\hat{H}$ (spectral radius $\times 1.3$ to 2.1 on unweighted graphs, $\times 1.9$ to 3.6 on weighted random ones, 9 graphs), but it inflates $\hat{b}$ as well, and the gauge-invariant ratio $\rho(H)$ decreased on 5 of the 9 graphs (`brock200_1` 57 to 37, `g500_50_0` 88 to 40) and increased on 3 (`p_hat300-3` 23 to 40). Iterated anchoring does not compound: each pass rebuilds the standard wrapper and anchors once, and $\lambda_{\max}$ even fell as the anchor improved (`g2000_95_0` weighted: 127 standard, then 407, 266, 266, 261).

4.3 Cold, warm and iterated anchoring

Cold anchoring loses: DIMACS 2 better and 9 worse (`gen400_p0.9_65` 65 $\to$ 52), weighted random graphs 15 better and 67 worse. Neither $\theta = 0.5$, nor a shuffled anchor (11/76), nor iterating (31/50) rescued it. By Corollary 3.6, anchoring makes the neighbourhood of Q attractive; around a poor clique that is a bad place.

Warm anchoring never lost on 366 graphs. It improved 3 DIMACS graphs (`keller5` 26 $\to$ 27, `gen400_p0.9_55` 53 $\to$ 55, `MANN_a45` 342 $\to$ 343), 8 of the weighted-56, 15 of 120 unweighted and 55 of 120 weighted random graphs, at 1.5 to 1.9 times the time of one pass. A second pass on a randomly perturbed wrapper of comparable cost improved 2, 0, 2 and 21 of the same.

Iterating it (ancwp) keeps climbing on dense weighted graphs: 35 of the 120 weighted random graphs improved beyond the first anchored pass (total weight +179, seven still improving at the cap of four passes), 3 of the weighted-56 (`w_nrm_200_90_1`: 263 $\to$ 267 $\to$ 268 $\to$ 270 $\to$ 272, the optimum), 2 unweighted random graphs and no DIMACS graph; 38 of the 70 proven weighted optima were reached. On the largest graphs iterated warm anchoring improved `keller6` 53 $\to$ 55 and `MANN_a81` 1096 $\to$ 1098, 3 of 12 unweighted $n = 2000$ graphs (`g2000_50_1` 15 $\to$ 16) and 8 of 12 weighted ones, with the largest gains seen (`g2000_95_0` 891 $\to$ 932, `g2000_90_0` 530 $\to$ 555), at 2 to 4 times the time.

The gains are local: a median of 2 vertices dropped and 3 added, 95–97% of the anchor kept (91% after iterating), against about 59% for the gains of a random wrapper. Whether meeting (1) exactly matters depends on density: on weighted random graphs with $p \leq 0.7$ the shuffled control did as well (11 improved against 7; 37 proven optima each), but for $p \in \{0.9, 0.95\}$ only the exact anchor worked (38 against 6), and likewise on dense unweighted graphs (12 against 1).

4.4 Blending in one step of $\lambda_{\max}$ minimization

Minimizing $\lambda_{\max}(H)$ over $\mathcal{H}(G, w)$ is the Lovász program (2); its optima are known to be highly degenerate, and Sections 5–6 explain why. One step of it was tried instead of the full minimization. The top eigenvector u of H was within about a percent of $z/\|z\|$, so the gradient

$2u_iu_j$ has cosine 0.98 to 1.00 with the gauge direction z_iz_j of Lemma 2.5, which changes nothing. The remainder drives $\hat{b}$ towards 0: by definition $\hat{b} = PHz/W(V)$ vanishes iff z is an eigenvector of H. The ratio $\rho(H)$ grew by a median factor of 1.23 to 1.65 per suite, where the shuffled control stayed at 1.00.

Against one warm anchored pass the step (icel) raised the proven weighted optima from 37 to 41 of 70, all four gains at $p \leq 0.5$ (5 better and 0 worse there, against 0 and 1 for icels), and the unweighted ones from 43 to 44 of 50; on dense graphs it was 7 better and 9 worse, no different from its control, and on DIMACS it gave back `MANN_a45` 343 → 342. The projected variant (ices) changed little (7 better, 5 worse on weighted random graphs). Without anchoring (lz, lzw) the step is mostly the effect of any large change of wrapper (Table 2). The step also raised the number of eigenvalues above μ_Q at the anchored point on 93 of the 120 weighted random graphs and lowered it on none (median 29.5 to 43): the benefit of icel over ancw is not a more stable anchor.

variant	proven optima (of 70)	vs head better/worse	$p \leq 0.5$	$p = 0.7$	$p \geq 0.8$
head	33	–	–	–	–
ancw	37	55/0	3/0	4/0	48/0
icel	41	59/0	7/0	4/0	48/0
lz	35	33/36	6/6	4/9	23/21
lzw	40	34/0	6/0	5/0	23/0
lzws	35	28/0	3/0	3/0	22/0
icel vs ancw		12/9	5/0	0/0	7/9
lzw vs lzws		20/18	5/1	3/3	12/14

Table 1: Weighted random graphs (120, of which 70 with proven optima). Time relative to head: ancw 1.88, icel 2.20, lz 1.34, lzw 1.98, lzws 1.76.

variant	DIMACS (70)	unweighted random (120)	weighted random (120)
lz	2/0	10/3	33/36
lzw	1/0	11/0	34/0
lzws	3/0	7/0	28/0

Table 2: One $\lambda_{\max}$ step without anchoring, better/worse against head. The DIMACS changes were `DSJC1000.5` 14 → 15 (lz, lzw), `gen400_p0.9_55` 53 → 54 (lz) and → 55 (lzws), `keller5` 26 → 27 and `MANN_a45` 342 → 343 (lzws). On the unweighted random graphs all three reach 43 of the 50 proven optima (head 42). For comparison, as second passes: a random wrapper 2/2/21, warm anchoring 3/15/55.

5 Anchoring several cliques

We say that H anchors the maximal cliques $Q_1, \ldots, Q_k$ at levels $\mu_1, \ldots, \mu_k$ if it anchors each Q_a at μ_a. By Lemma 3.2 this is a system of linear equations in the defects, one for each pair

(Q_a, i) with $i \notin Q_a$:

$$\sum_{j \in Q_a \setminus N(i)} w_j \tau_{ij} = \mu_a. \tag{7}$$

A non-edge $\{i, j\}$ can now occur in several equations, and the system can be inconsistent. For $Q_a \neq Q_b$ we write $D_a = Q_a \setminus Q_b$, $D_b = Q_b \setminus Q_a$ and $K = Q_a \cap Q_b$; for maximal cliques D_a and D_b are nonempty.

Theorem 5.1 (level identity)**.** *If H anchors two distinct maximal cliques Q_a, Q_b at levels μ_a, μ_b, then*

$$\mu_a W(Q_b \setminus Q_a) = \mu_b W(Q_a \setminus Q_b).$$

Proof. A vertex $i \in D_b$ lies outside Q_a and is adjacent to all of K, so its non-neighbours in Q_a lie in D_a and (7) reads $\sum_{j \in D_a \setminus N(i)} w_j \tau_{ij} = \mu_a$. Multiplying by w_i and summing over $i \in D_b$ gives $\sum w_i w_j \tau_{ij} = \mu_a W(D_b)$, the sum running over the non-edges between D_a and D_b. The same computation from the side of Q_b gives the same sum equal to $\mu_b W(D_a)$. □

A second proof: apply Proposition 3.5 to $u = x^{Q_b} - x^{Q_a}$ from both ends. The Rayleigh quotient of u is $\mu_a(1 + (W_b - W_a)/W(\triangle)) = \mu_b(1 + (W_a - W_b)/W(\triangle))$, with $W_a = W(Q_a)$ and $W(\triangle) = W(D_a) + W(D_b)$, which rearranges to the identity.

Corollary 5.2. *Cliques of equal weight anchored in one wrapper share one level; in particular all maximum weight cliques do. Pairwise disjoint anchored cliques have levels proportional to their weights.*

For three or more cliques the pairwise ratios $W(D_a)/W(D_b)$ must also be consistent around every cycle, which for overlapping cliques of different weights generically fails, and then only the level 0 remains.

Theorem 5.3 (collision)**.** *Let H anchor distinct maximal cliques $Q_1, \ldots, Q_k$, $k \geq 2$, at a common level $\mu \neq 0$, and let $E = \ker(\mu I - \hat{H}) \cap z^{\perp}$. Then*

- *(a) every difference $x^{Q_a} - x^{Q_b}$ lies in E, so μ is an eigenvalue of $\hat{H}$ of multiplicity at least $d = \dim \operatorname{aff}\{x^{Q_1}, \ldots, x^{Q_k}\} \geq 1$;*
- *(b) $\hat{b} \perp E$: the linear form c vanishes on the whole eigenspace;*
- *(c) $\hat{x}^{Q_a} = \hat{x}_p + e_a$ with a common $\hat{x}_p \in E^{\perp}$ and $e_a \in E$; if the $W(Q_a)$ are equal, no x^{Q_a} lies on the secular curve, which passes through $x^0 + \hat{x}_p$ at μ.*

Proof. By Lemma 3.3, $(\mu I - \hat{H})\hat{x}^{Q_a} = \hat{b}$ for every a; subtracting two of these gives (a). Then $\hat{b}$ lies in the range of the symmetric operator $\mu I - \hat{H}$ on $z^{\perp}$, which is $E^{\perp}$, giving (b). Decompose $\hat{x}^{Q_a}$ along $E^{\perp} \oplus E$; the $E^{\perp}$ components solve the same nonsingular system on $E^{\perp}$ and so coincide. If the weights are equal, $\|\hat{x}^{Q_a}\|^2 = 1/W(Q_a) - 1/W(V)$ is the same for all a, hence so is $\|e_a\|$; if one e_a vanished all would, and all the indicators would coincide. Since $c = 0$ on E, the secular curve has no pole at μ and tends to $x^0 + (\mu I - \hat{H})^{+}\hat{b} = x^0 + \hat{x}_p$ there; at any other multiplier it cannot meet x^{Q_a}, whose multiplier is μ by Lemma 3.3. □

This is precisely the "degenerate case of multiple eigenvalues with $c_i = 0$" that [CW, Section 5] sets aside: anchoring several cliques of equal weight forces it, whatever the remaining free entries. The contrapositive is useful as well: if $\hat{b}$ has a nonzero component along $x^{Q_a} - x^{Q_b}$ for two cliques of equal weight, then H does not anchor both. Making a degenerate direction "active" ($c \neq 0$ on it) and anchoring the cliques that span it are mutually exclusive.

Theorem 5.4 (two cliques)**.** *Let $Q_a \neq Q_b$ be maximal cliques and let B be the bipartite graph on $D_a \cup D_b$ whose edges are the non-adjacent pairs of G between D_a and D_b. There is $H \in \mathcal{H}(G, w)$ anchoring Q_a and Q_b at nonzero levels iff*

(i) every connected component Γ of B satisfies $W(\Gamma \cap D_a)/W(\Gamma \cap D_b) = W(D_a)/W(D_b)$, and

(ii) if some vertex outside $Q_a \cup Q_b$ is adjacent to all of $Q_a \triangle Q_b$, then $W(Q_a) = W(Q_b)$.

The levels then satisfy $\mu_a : \mu_b = W(D_a) : W(D_b)$.

Proof. Every vertex of B has a neighbour in B: a vertex $j \in D_a$ adjacent to all of D_b would be adjacent to all of Q_b, contradicting maximality. The equations (7) and the defects they involve split into independent blocks.

Outside vertices. For $i \notin Q_a \cup Q_b$ let $A_i = D_a \setminus N(i)$, $B_i = D_b \setminus N(i)$, $K_i = K \setminus N(i)$. The equations of i are $\tau(K_i) + \tau(A_i) = \mu_a$ and $\tau(K_i) + \tau(B_i) = \mu_b$, where $\tau(S) = \sum_{j \in S} w_j \tau_{ij}$, and their defects occur in no other equation. By maximality $K_i \cup A_i$ and $K_i \cup B_i$ are nonempty. If $A_i = B_i = \emptyset$ (i.e. i is adjacent to all of $Q_a \triangle Q_b$), both read $\tau(K_i) = \mu_a = \mu_b$; otherwise they are solvable for any levels (if, say, $B_i = \emptyset \neq A_i$, set $\tau(K_i) = \mu_b$ and $\tau(A_i) = \mu_a - \mu_b$).

The two differences. As in the proof of Theorem 5.1 the remaining equations are, with $\xi_{ij} = w_i w_j \tau_{ij}$ on the edges of B, $\sum_j \xi_{ij} = w_i \mu_a$ for $i \in D_b$ and $\sum_i \xi_{ij} = w_j \mu_b$ for $j \in D_a$: edge values of B with prescribed sums at the vertices. The incidence matrix of a connected bipartite graph has corank one, its left kernel spanned by the vector that is $+1$ on one side and -1 on the other; so this is solvable iff every component satisfies $\mu_a W(\Gamma \cap D_b) = \mu_b W(\Gamma \cap D_a)$. Summing over components gives Theorem 5.1, and for nonzero levels the component conditions hold iff (i) does. With (ii) for the outside vertices this proves the claim. □

Corollary 5.5. *Any two maximum weight cliques can be anchored in one wrapper at a common positive level with all defects nonnegative.*

Proof. Here $W(D_a) = W(D_b)$ and the level is common. For $S \subseteq D_b$ let $N_B(S)$ be its neighbourhood in B. The set $(Q_a \setminus N_B(S)) \cup S$ is a clique (vertices of S are adjacent to K, and to every vertex of D_a outside $N_B(S)$), of weight $W(Q_a) - W(N_B(S)) + W(S)$, so $W(N_B(S)) \geq W(S)$ by maximality of $W(Q_a)$. By the supply–demand theorem of Gale [Ga57], the transportation problem with supplies $w_i \mu$ on D_b, demands $w_j \mu$ on D_a and routes along the edges of B then has a nonnegative solution. For an outside vertex i take $\tau(K_i) = \mu$ and $\tau(A_i) = \tau(B_i) = 0$ if A_i or B_i is empty (K_i is then nonempty), and $\tau(K_i) = 0$, $\tau(A_i) = \tau(B_i) = \mu$ otherwise, spreading each sum evenly over its non-edges. □

Nonnegative, however, does not mean proper:

Proposition 5.6 (forced blindness)**.** *Let H anchor Q_a and Q_b at a common level, and let $i \notin Q_a \cup Q_b$ be adjacent to all of D_b. Then $\sum_{j \in D_a \setminus N(i)} w_j \tau_{ij} = 0$. In particular a proper wrapper cannot anchor both if such an i has a non-neighbour in D_a; with nonnegative defects all these τ_{ij} vanish, so $h_{ij} = z_i z_j$ on those non-edges: the wrapper does not distinguish them from edges.*

Proof. Subtract the two equations of i in the proof of Theorem 5.4: with $B_i = \emptyset$ they give $\tau(A_i) = \mu_a - \mu_b = 0$. □

In dense graphs, maximum cliques that differ by a single exchange $Q_b = Q_a - j + j'$ are common, and every outside vertex i adjacent to j' but not to j then makes the non-edge $\{i, j\}$ invisible, here without any sign condition, since $D_a = \{j\}$ leaves a single defect in the sum. Once invisible non-edges cover all the non-neighbours of some vertex in some other anchored clique, that vertex's equation reads $0 = \mu$ and the common level collapses; Example 6.9 shows this with three maximum cliques.

6 Anchoring every maximum clique

6.1 The best relaxation is the Lovász number

Theorem 6.1. *For every $H \in \mathcal{H}(G, w)$, $v(H) \le 1/\vartheta(\bar{G}, w)$, with equality whenever $\lambda_{\max}(H) = \vartheta(\bar{G}, w)$. Hence*

$$\max_{H \in \mathcal{H}(G,w)} \min\{x^T x : \ x \in \mathcal{W}^H\} = \frac{1}{\vartheta(\bar{G}, w)}.$$

Proof. Equality for Lovász-optimal H. For $x \in \mathcal{W}^H$, $1 = x^T H x \le \lambda_{\max}(H)\, x^T x = \vartheta(\bar{G}, w)\, x^T x$, so $v(H) \ge 1/\vartheta(\bar{G}, w)$; the upper bound below then gives equality.

The upper bound. Consider the semidefinite relaxation

$$v_{\mathrm{SDP}}(H) = \min\{\operatorname{tr} X : \ \langle H, X\rangle = 1, \ \langle zz^T, X\rangle = 1, \ X \succeq 0\}.$$

If $x \in \mathcal{W}^H$ then $X = xx^T$ is feasible with $\operatorname{tr} X = x^T x$, so $v_{\mathrm{SDP}}(H) \le v(H)$. Conversely, the feasible set is nonempty and the trace is bounded below and coercive on it, so the minimum is attained and the optimal set is a nonempty compact face of the feasible set. An extreme point of it is an extreme point of the feasible set, which is cut out of the positive semidefinite cone by two linear equations; by the rank bound of Barvinok and Pataki [Ba95, Pa98] it has rank r with $r(r+1)/2 \le 2$, so $r \le 1$, and $r = 0$ is excluded by $\langle zz^T, X\rangle = 1$. Writing it as xx^T with $z^T x = 1$ (replacing x by $-x$ if necessary) gives $x \in \mathcal{W}^H$ with $x^T x = v_{\mathrm{SDP}}(H)$. Hence $v(H) = v_{\mathrm{SDP}}(H)$.

Now let X^* be a Lovász optimum. H and zz^T agree on the support of X^* (the diagonal and E), so $X = X^*/\vartheta(\bar{G}, w)$ has $\langle H, X\rangle = \langle zz^T, X\rangle = 1$ and $\operatorname{tr} X = 1/\vartheta(\bar{G}, w)$. Thus $v(H) = v_{\mathrm{SDP}}(H) \le 1/\vartheta(\bar{G}, w)$. □

So every wrapping matrix gives an upper bound $1/v(H) \ge \vartheta(\bar{G}, w) \ge \omega(G, w)$ on the weighted clique number, and the Lovász-optimal wrappers give the best of them. In the sense of the relaxed least squares program they are the best wrappers. By Lemma 2.5, v is constant on gauge orbits. For the standard wrapper of a regular graph the bound is a classical one:

Corollary 6.2. *Let G be d-regular, not complete, with unit weights, and let λ_2 be the largest eigenvalue of A_G on $\mathbf{1}^\perp$. Then*

$$\frac{1}{v(H_0)} = \frac{n(1 + \lambda_2)}{n - d + \lambda_2},$$

Hoffman's ratio bound [Ho70] on $\alpha(\bar{G}) = \omega(G)$ applied to the $(n - 1 - d)$-regular graph $\bar{G}$.

Proof. Here $x^0 = \mathbf{1}/n$ and $H_0 x^0 = (1+d)x^0$, so $\hat{b} = 0$ and $s = 1-(1+d)/n$. As A_G preserves $\mathbf{1}^\perp$, $\lambda_{\max}(\hat{H}_0) = 1+\lambda_2$, which is at least 1 for a non-complete graph (λ_2 is then the second eigenvalue of A_G, at least 0 by interlacing with two non-adjacent vertices). The program is $\min \|\hat{x}\|^2$ subject to $\hat{x}^T \hat{H}_0 \hat{x} = s$, with value $s/(1+\lambda_2)$, so $v(H_0) = 1/n + (n-1-d)/(n(1+\lambda_2))$, which rearranges to the claim. The smallest eigenvalue of $A_{\bar{G}}$ is $-1-\lambda_2$, and Hoffman's bound for $\bar{G}$ is $n(1+\lambda_2)/(n-1-d+1+\lambda_2)$. □

With Theorem 6.1 this recovers Lovász's inequality $\vartheta \le$ Hoffman's bound for regular graphs [Lo79, Theorem 9]: passing from the standard to a Lovász-optimal wrapper improves the relaxed bound from Hoffman's to ϑ.

6.2 Global anchoring

Theorem 6.3. *For a clique Q the following are equivalent:*

(a) x^Q is a global minimizer of (5) for some $H \in \mathcal{H}(G,w)$;

(b) $W(Q) = \omega(G,w) = \vartheta(\bar{G},w)$.

If they hold, every Lovász-optimal H anchors every maximum weight clique at level $\vartheta(\bar{G},w)$ ($C = 0$), and every maximum weight clique indicator is a global minimizer of (5) for it. Moreover, if some H anchors a clique Q at a level $\mu > 0$ with $\mu \ge \lambda_{\max}(\hat{H})$, then (b) holds.

Proof. (a)⇒(b): $1/W(Q) = \|x^Q\|^2 = v(H) \le 1/\vartheta(\bar{G},w)$ by Theorem 6.1, so $W(Q) \ge \vartheta(\bar{G},w) \ge \omega(G,w) \ge W(Q)$. (b)⇒(a): for Lovász-optimal H, $v(H) = 1/\vartheta(\bar{G},w) = 1/W(Q) = \|x^Q\|^2$. For a Lovász-optimal H and a maximum weight clique Q', the Rayleigh quotient of $z_{Q'}$ is $W(Q') = \vartheta(\bar{G},w) = \lambda_{\max}(H)$ (Lemma 2.2), so $z_{Q'}$ is a top eigenvector, $Hz_{Q'} = \vartheta(\bar{G},w) z_{Q'}$, which is Lemma 3.2(c) with $\mu = W(Q') = \vartheta(\bar{G},w)$. The last claim follows from Lemmas 3.3 and 2.3(a) and (a)⇒(b). □

So the relaxed program can be exact, in the sense that some wrapper makes a clique indicator its global minimizer, exactly when $\vartheta = \omega$; and then *anchoring one maximum clique globally is the same as anchoring all of them.* On graphs with $\vartheta > \omega$, which include most hard benchmark graphs, every anchored clique is a non-global stationary point. This is why QUALEX-MS must examine the stationary points in all intervals of the secular equation and not only the one beyond $\lambda_{\max}$.

Corollary 6.4 (the hard case)**.** *Let $\vartheta(\bar{G},w) = \omega(G,w)$ and let G have $k \ge 2$ maximum weight cliques, whose indicators span an affine space of dimension d. For every Lovász-optimal H, $\mu = \vartheta(\bar{G},w)$ equals $\lambda_{\max}(\hat{H})$, with multiplicity at least d and with $\hat{b}$ orthogonal to its eigenspace. The multiplicity of $\lambda_{\max}(H)$ is at least the rank of every Lovász optimum, in particular at least* $\dim \operatorname{span}\{z_Q\}$ *over the maximum weight cliques Q.*

Proof. All maximum cliques are anchored at $\mu = \vartheta(\bar{G},w)$ (Theorem 6.3). The restriction of $\hat{H}$ to $z^\perp$ is a compression of H, so $\lambda_{\max}(\hat{H}) \le \lambda_{\max}(H) = \vartheta(\bar{G},w)$ by interlacing, while Theorem 5.3 makes μ an eigenvalue of $\hat{H}$ of multiplicity at least d with $\hat{b}$ orthogonal to its eigenspace. By (4) the range of every Lovász optimum X lies in $\ker(\vartheta(\bar{G},w)I - H)$, and the matrices X_Q of Lemma 2.2 are Lovász optima when $W(Q) = \vartheta(\bar{G},w)$. □

In trust region terms, the sphere of radius $\|\hat{x}^Q\|$ is in the *hard case* [MS83]: its global maximizers form a sphere inside $x^0 + \hat{x}_p + E$, which contains all the maximum clique indicators, none of them on the secular curve. This is the structural reason why the Lovász-optimal wrappers, although best by Theorem 6.1, are useless for recognizing a clique through the secular equation when there are several maximum cliques. When $\vartheta > \omega$ the Lovász-optimal wrappers are degenerate for a different reason, the generic multiplicity of optimal eigenvalues [Ov92, Pa98]; Section 7 shows both.

6.3 Colourings, holes and perfect graphs

In this subsection all weights are 1, so $z = \mathbf{1}$ and $\hat{H}$, τ are those of an unweighted graph.

Proposition 6.5 (the colouring wrapper)**.** *Let $\chi(G) = \omega(G) = \omega$, with colour classes $S_1, \dots, S_\omega$. Then*

$$H_\chi = \omega I - \Big(\omega \sum_{c=1}^{\omega} \mathbf{1}_{S_c}\mathbf{1}_{S_c}^T - \mathbf{1}\mathbf{1}^T\Big)$$

lies in $\mathcal{H}(G, \mathbf{1})$, with defects $\tau_{ij} = \omega$ inside colour classes and $\tau_{ij} = 0$ between them, has $\lambda_{\max}(H_\chi) = \omega$ and anchors every maximum clique at level ω ($C = 0$).

Proof. The diagonal of H_χ is $\omega - (\omega - 1) = 1$; between classes the entries are 1, which covers all edges; inside a class (non-adjacent pairs) they are $1 - \omega$. The bracket is positive semidefinite by the Cauchy–Schwarz inequality, $\omega \sum_c (\mathbf{1}_{S_c}^T x)^2 \geq (\sum_c \mathbf{1}_{S_c}^T x)^2$, so $H_\chi \preceq \omega I$, and $\lambda_{\max}(H_\chi) = \omega$ by Lemma 2.2. A maximum clique Q meets every class in one vertex; for $i \notin Q$ in class c, $\sum_{j \in Q} h_{ij} = (1 - \omega) + (\omega - 1) = 0$, so $C = 0$. □

The colouring wrapper sees only the complete ω-partite graph on the colour classes: the indicator of every transversal of the colouring lies on $\mathcal{W}^{H_\chi}$ and has norm $1/\sqrt{\omega}$, exactly like a maximum clique, although most transversals are not cliques. Such pseudo-cliques are what the strict inequality of [CW, Proposition 2] rules out; the colouring wrapper is improper as soon as two colour classes are not completely joined.

Theorem 6.6. *Let G be an odd hole C_{2k+1} or an odd antihole $\bar{C}_{2k+1}$, $k \geq 2$, with unit weights. Every $H \in \mathcal{H}(G, \mathbf{1})$ that anchors all maximum cliques of G does so at level 0, and then $H = \mathbf{1}\mathbf{1}^T$.*

Proof. All maximum cliques share one level μ (Corollary 5.2). Vertices are taken modulo $n = 2k + 1$.

Odd holes. The maximum cliques are the edges $\{a, a + 1\}$. For $Q = \{a, a + 1\}$ the outside vertex $a + 2$ misses only a and $a - 1$ misses only $a + 1$, so every non-edge at distance 2 has $\tau = \mu$; a vertex v with $a + 3 \leq v \leq a - 2$ misses both, giving $\tau_{a,v} + \tau_{a+1,v} = \mu$. With $v = a + j + 1$ this reads $\tau_{a,a+j+1} = \mu - \tau_{a+1,a+j+1}$, so by induction on j every pair $\{b, b + j\}$ has $\tau = \mu$ for even j and $\tau = 0$ for odd j, $2 \leq j \leq n - 2$. The pair $\{b, b + 2\}$ is also the pair $\{b + 2, (b + 2) + (n - 2)\}$ with $n - 2$ odd, so $\mu = 0$, and then every defect is 0.

Odd antiholes. The non-edges are the pairs $\{b, b+1\}$; write τ_b for their defects. The maximum cliques of $\bar{C}_{2k+1}$ are the stable sets of size k of C_{2k+1}, namely $Q_a = \{a, a + 2, \dots, a + 2k - 2\}$ (consecutive elements differ by 2 except for one gap of 3). The non-neighbours of an outside vertex v in Q_a are its neighbours on the cycle that belong to Q_a. For $v = a + 2k - 1$ this is $\{a + 2k - 2\}$ and for $v = a + 2k$ it is $\{a\}$, so $\tau_{a+2k-2} = \mu$ and $\tau_{a-1} = \mu$ for all a: every defect equals μ. For $v = a + 1$ (which is outside Q_a since $k \geq 2$) it is $\{a, a + 2\}$, giving $\tau_a + \tau_{a+1} = \mu$, i.e. $2\mu = \mu$. So $\mu = 0$ and all defects vanish. □

Even holes behave differently. The same induction shows that H anchors all maximum cliques of C_{2k} ($k \geq 2$) iff $\tau = \mu$ at even distances and $\tau = 0$ at odd distances, which is consistent because j and $2k - j$ have the same parity. At $\mu = 2$ this is the colouring wrapper of the bipartition, so it is Lovász-optimal ($\vartheta = \omega = 2$). For $k \geq 3$ the odd-distance non-edges are invisible to *every* wrapper anchoring all maximum cliques, hence to every Lovász-optimal wrapper. For C_6 the three long diagonals $\{i, i+3\}$ then satisfy $x^T H x = 1$ for $x = (e_i + e_{i+3})/2 \in \Delta_n$, which has $x^T x = 1/2 = 1/\omega$: they are global minimizers of the program of [CW, Theorem 1] although they are not cliques. The Lovász-optimal wrappers of C_6 are necessarily improper.

Theorem 6.7. *For a graph G with unit weights the following are equivalent:*

(a) G is perfect;

(b) for every induced subgraph G' of G some $H \in \mathcal{H}(G', \mathbf{1})$ anchors all maximum cliques of G' at a common nonzero level.

When (a) holds, H in (b) can be taken Lovász-optimal for G'.

Proof. (a)⇒(b): induced subgraphs of perfect graphs are perfect, so $\chi(G') = \omega(G')$, and Proposition 6.5 applies. (b)⇒(a): if G is not perfect, by the strong perfect graph theorem [CRST06] it has an induced odd hole or odd antihole, for which (b) fails by Theorem 6.6. □

Condition (b) is linear (no semidefiniteness is involved), but it must be required hereditarily: an imperfect graph with a unique maximum clique satisfies it for $G' = G$, and so do the graphs of Section 7 with $\vartheta > \omega$ whose maximum cliques can all be anchored. One direction holds without heredity:

Corollary 6.8. *If the maximum weight cliques of G cannot all be anchored at a common nonzero level, then $\vartheta(\bar{G}, w) > \omega(G, w)$.*

Proof. If $\vartheta(\bar{G}, w) = \omega(G, w)$, a Lovász-optimal wrapper anchors them all at level $\vartheta(\bar{G}, w)$ (Theorem 6.3). □

Example 6.9. The unweighted random graph `g200_80_0` ($n = 200$, $p = 0.8$, $\omega = 26$) has three maximum cliques. $Q_2 = Q_1 - 14 + 53$ (vertex labels of the DIMACS file), and Q_3 shares only four vertices with them. The vertices 95, 113, 158 and 176 of Q_3 are adjacent to 53 but not to 14, so by Proposition 5.6 (with $D_a = \{14\}$, a single vertex, so without any sign condition) $\tau_{i,14} = 0$ for each of them. But these four are exactly the non-neighbours of vertex 14 in Q_3, so the equation of 14 for Q_3 reads $0 = \mu$. The system has 418 distinct equations in 1693 defects and is still inconsistent; this certificate combines nine of them.

6.4 Quasigroup completion: why the hard case does not help

Busygin and Pasechnik [BP03] reduce the quasigroup completion problem (QCP) to the maximum independent set problem. For a partial latin square of order n with h empty cells (holes), the *QCP graph* G has as vertices the triples (i, j, k) such that (i, j) is a hole and the symbol k occurs neither in row i nor in column j, and two triples are adjacent when they agree in two coordinates; it is the subgraph of the Hamming graph $H(3, n)$ induced by the non-neighbours of the prefilled triples. Its independent sets have at most h vertices, and the independent sets of size h are exactly the completions of the square [BP03, Lemma 2]. The candidates of one hole

form a clique, so $\overline{\chi}(G) \le h$ [BP03, Lemma 3], and deciding whether $\alpha(G) = \overline{\chi}(G)$ is NP-hard even when this clique partition is given [BP03, Theorem 1]. Hence every completable instance has

$$\alpha(G) = \vartheta(G) = \overline{\chi}(G) = h,$$

with $\vartheta(G)$ the bound (2) for the graph $\bar{G}$, and the hard instances of QCP are among these or look like them to ϑ. The independent sets of G are the cliques of $\bar{G}$, so we use the wrapping matrices of $\bar{G}$ with unit weights; defects now sit on the edges of G, and the standard wrapping matrix of $\bar{G}$ is $J - A_G$.

Besides the cells, G has two more partitions into cliques: the row–symbol lines $\{(i, \cdot, k)\}$ and the column–symbol lines $\{(\cdot, j, k)\}$. Two distinct triples share at most one line (sharing two coordinates of two different kinds would make them equal).

Proposition 6.10. *Let G be the QCP graph of a completable instance with h holes, and let $\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3$ be its partitions into cells, row–symbol lines and column–symbol lines, each of h cliques. Then:*

(a) $H = J - \frac{h}{3} A_G$, the standard wrapping matrix of $\bar{G}$ up to gauge, has $\lambda_{\max}(H) = h = \vartheta(G)$; it is Lovász-optimal and anchors every completion S at level h ($C = 0$);

(b) its eigenspace for h is $L = \{x : \ \mathbf{1}_C^T x$ is the same for all $C \in \mathcal{P}_t,\ t = 1, 2, 3\}$;

(c) the global minimizers of (5) for H are the $x \in L$ with $\mathbf{1}^T x = 1$ and $x^T x = 1/h$, and the nonnegative ones among them are exactly the indicators $\mathbf{1}_S/h$ of the completions S.

Proof. For $t = 1, 2, 3$ let $K_t = h \sum_{C \in \mathcal{P}_t} \mathbf{1}_C \mathbf{1}_C^T - J$, which is positive semidefinite by the Cauchy–Schwarz inequality as in Proposition 6.5, and $H_t = hI - K_t$, the colouring wrapper of $\bar{G}$ for the colouring $\mathcal{P}_t$. H_t has $1 - h$ on the pairs that share a line of $\mathcal{P}_t$ and 1 elsewhere. Every edge of G shares exactly one line, so $(H_1 + H_2 + H_3)/3$ has 1 on the diagonal and on the non-adjacent pairs of G and $1 - h/3$ on its edges: it is H, with defect $h/3$ on every non-edge of $\bar{G}$, i.e. $(J - A_G)_t$ with $1 + t = h/3$ in the notation of Lemma 2.5. Moreover $H = hI - \frac{1}{3}(K_1 + K_2 + K_3) \preceq hI$, and $\ker(hI - H) = \bigcap_t \ker K_t$, where $\ker K_t$ consists of the vectors whose sums over the h classes of $\mathcal{P}_t$ are equal (the equality case of the Cauchy–Schwarz inequality). This is (b). A completion meets every line exactly once, so $\mathbf{1}_S \in L$ and $H\mathbf{1}_S = h\mathbf{1}_S$, which gives $\lambda_{\max}(H) = h$, the anchoring at level h (Lemma 3.2(c)), and $\vartheta(G) \le h = \alpha(G) \le \vartheta(G)$.

For (c), $v(H) = 1/h$ by Theorem 6.1. A point of $\mathcal{W}^H$ has $1 = x^T H x \le h\, x^T x$, with equality iff $x \in L$; so the global minimizers are the points of L with $\mathbf{1}^T x = 1$ and $x^T x = 1/h$ (on L, $x^T H x = h\, x^T x$). Let such an x be nonnegative. Each partition has h classes covering V, so all its class sums equal $1/h$, and

$$x^T x = \sum_{\text{holes}} \sum_{k} x_{ijk}^2 \ \le \ \sum_{\text{holes}} \Big(\sum_{k} x_{ijk} \Big)^2 = \frac{1}{h},$$

with equality iff every hole carries a single candidate, with value $1/h$. Those candidates meet every row–symbol and column–symbol line once, so they form a completion. □

So the standard wrapper that QUALEX-MS builds for the independent sets of a completable QCP graph is Lovász-optimal, its trust region program at the radius of the completions is in the hard case with the degenerate eigenspace $L \cap \mathbf{1}^\perp$, and the completions are exactly the points

of the sphere of global minimizers that lie in the nonnegative orthant, i.e. the points of largest norm of the polytope $\{x \in L : \ x \geq 0, \ \mathbf{1}^T x = 1\}$ of fractional completions. The hard case therefore carries all the information about the completions, but finding a nonnegative point on that sphere is QCP itself, which is NP-complete [Co84]. The colouring wrapper of the cells alone (Proposition 6.5 for $\mathcal{P}_1$) is Lovász-optimal as well but blind to every row and column conflict: by the same argument every transversal, one candidate for each hole, is a global minimizer, and its top eigenspace has dimension $|V| - h + 1$. For the empty square, $G = H(3, n)$, L is the sum $V_0 \oplus V_3$ of the eigenspaces of the Hamming scheme, of dimension $(n-1)^3 + 1$, and (a) is the tightness of Hoffman's bound $\alpha(H(3,n)) = n^2$ (Corollary 6.2).

Every Lovász-optimal wrapper has at least the rank of every Lovász optimum as the multiplicity of its top eigenvalue (Corollary 6.4), and that rank is at least the dimension spanned by the completions. Section 7.6 measures where between that span and $\dim L$ the Lovász optimum falls on random instances.

6.5 SAT01: exact covers with contradictions

Busygin's SAT01 [Bu07] puts QCP into a general weighted setting. An instance SAT01(A, Γ) is a 0-1 matrix A of size $m \times n$ without zero columns and a graph Γ on the n variables, and it asks for $x \in \{0,1\}^n$ with $Ax = \mathbf{1}$ whose support is independent in Γ. We identify each equation with the set $e \subseteq V$ of its variables and write $\mathcal{E}$ for the m equations. Γ contains every pair of variables that share an equation and possibly further pairs; all its edges are called *contradictions*. With the weights $w = A^T\mathbf{1}$, the number of equations of each variable, an independent set of Γ meets every equation at most once, so its weight is at most m, with equality iff it is the support of a solution [Bu07, Lemma 1]. The equations are m cliques of Γ that cover every vertex j exactly w_j times, so $\vartheta(\Gamma, w) \leq m$ [Bu07, Theorem 1]. Many NP problems reduce to SAT01 in a natural way (CNF satisfiability, Hamilton cycles, graph and subgraph isomorphism, QCP, the 15-puzzle, the Hanoi tower, blocksworld planning, factoring). A satisfiable instance has $\alpha = \vartheta = m$, so $\vartheta < m$ refutes an instance in polynomial time, and only the unsatisfiable instances with $\alpha < \vartheta = m$ escape it.

In the notation of this note $G = \bar{\Gamma}$: two variables are adjacent iff they do not contradict, the solutions are the cliques of weight m (maximal cliques, [Bu07, Claim 1]), $\omega(G, w) = \alpha(\Gamma, w)$, $\vartheta(\bar{G}, w) = \vartheta(\Gamma, w)$, and the non-edges of G are the contradictions. The QCP graph of Section 6.4 is the instance with the cells, row–symbol lines and column–symbol lines as equations, all weights 3, $m = 3h$, and no contradictions besides the equations. Let $D = \operatorname{diag}(w)$, so that $z = D^{1/2}\mathbf{1}$, and $M = A^T A$: M_{ij} counts the equations shared by i and j, and $M_{jj} = w_j$.

Proposition 6.11. *Let*

$$H_A = zz^T + m\left(I - D^{-1/2} M D^{-1/2}\right).$$

(a) $H_A \in \mathcal{H}(G, w)$, with defect $\tau_{ij} = mM_{ij}/(w_i w_j)$ on every contradiction; in particular $\tau_{ij} = 0$ on the contradictions that share no equation.

(b) $mI - H_A = D^{-1/2} A^T (mI - \mathbf{1}\mathbf{1}^T) A D^{-1/2} \succeq 0$, so $\lambda_{\max}(H_A) \leq m$, and $\ker(mI - H_A) = L_A := D^{1/2}\{u : \ Au \in \mathbb{R}\mathbf{1}\}$.

(c) If $\vartheta(\bar{G}, w) = m$, H_A is Lovász-optimal; conversely, if $L_A \neq 0$ (e.g. when $n > m$) and H_A is Lovász-optimal, $\vartheta(\bar{G}, w) = m$. If the instance is satisfiable, $\vartheta(\bar{G}, w) = m$ and H_A anchors every solution S at level m ($C = 0$).

(d) If $\vartheta(\bar{G}, w) = m$, the global minimizers of (5) for H_A are the $x \in L_A$ with $z^T x = 1$ and $x^T x = 1/m$, and the nonnegative ones among them are exactly the indicators $x^S = z_S/m$ of the exact covers S of A (the 0-1 solutions of $Ax = \mathbf{1}$), whether or not they respect the contradictions outside the equations.

Proof. (a) As A is 0-1, $M_{jj} = w_j$ and the diagonal of H_A is $w_j + m(1-1) = w_j$. Off the diagonal $(H_A)_{ij} = z_i z_j - mM_{ij}/(z_i z_j) = z_i z_j\big(1 - mM_{ij}/(w_i w_j)\big)$. Adjacent vertices of G share no equation, so there $(H_A)_{ij} = z_i z_j$.

(b) As $D^{-1/2}A^T\mathbf{1} = D^{-1/2}w = z$,

$$mI - H_A = m\,D^{-1/2}MD^{-1/2} - zz^T = D^{-1/2}A^T\big(mI - \mathbf{1}\mathbf{1}^T\big)AD^{-1/2}.$$

Since $mI - \mathbf{1}\mathbf{1}^T$ is positive semidefinite with kernel $\mathbb{R}\mathbf{1}$, y lies in $\ker(mI - H_A)$ iff $AD^{-1/2}y \in \mathbb{R}\mathbf{1}$.

(c) By (2) and (b), $\vartheta(\bar{G}, w) \le \lambda_{\max}(H_A) \le m$, and $\lambda_{\max}(H_A) = m$ when $L_A \ne 0$; both implications follow. A solution S is a clique of weight m, so $\vartheta(\bar{G}, w) \ge m$ by Lemma 2.2, and $AD^{-1/2}z_S = A\mathbf{1}_S = \mathbf{1}$, so $z_S \in L_A$ and $H_A z_S = mz_S$, which is the anchoring at level $m = W(S)$ by Lemma 3.2(c).

(d) $\lambda_{\max}(H_A) = m = \vartheta(\bar{G}, w)$, so $v(H_A) = 1/m$ by Theorem 6.1, and a point of $\mathcal{W}^{H_A}$ has $1 = x^T H_A x \le m\,x^T x$, with equality iff $x \in L_A$; this gives the global minimizers. Write such an x as $D^{1/2}u$ with $Au = t\mathbf{1}$. Then $z^T x = w^T u = \mathbf{1}^T Au = tm$, so $t = 1/m$, and $p = mu$ satisfies $Ap = \mathbf{1}$. If $x \ge 0$ then $p \ge 0$, hence $p \le \mathbf{1}$, since every variable lies in an equation on which p sums to 1, and

$$m^2\,x^T x = \sum_j w_j p_j^2 \;\le\; \sum_j w_j p_j = \mathbf{1}^T Ap = m,$$

with equality iff $p \in \{0,1\}^n$. So $x^T x = 1/m$ iff $p = \mathbf{1}_S$ for an exact cover S, and then $x = D^{1/2}\mathbf{1}_S/m = x^S$. □

For QCP every pair shares at most one line, so $M = 3I + A_\Gamma$ and $H_A = 3J + 3hI - h(3I + A_\Gamma) = 3\big(J - \frac{h}{3}A_\Gamma\big)$, three times the wrapper of Proposition 6.10 (whose weights are 1), with $L_A = L$; that proposition is the case of (a)–(d) in which every exact cover is a solution. In general H_A sees only the equations: it is blind to every contradiction that shares no equation, and the nonnegative points of its sphere of global minimizers include the exact covers that violate such contradictions. It is always one of the Lovász-optimal wrappers when $\vartheta = m$, but on an unsatisfiable instance with $\vartheta < m$ it is not. Which instances these are is decided by the following relaxation of the solutions.

Theorem 6.12. *$\vartheta(\bar{G}, w) = m$ iff SAT01(A, Γ) has a* vector solution*: vectors $q_1, \ldots, q_n$ in some $\mathbb{R}^r$ and a unit vector d with*

$$q_i^T q_j = 0 \text{ for every contradiction } \{i,j\}, \qquad \sum_{j \in e} q_j = d \text{ for every equation } e \in \mathcal{E}.$$

The Lovász optima are then exactly the matrices $X = \big(z_i z_j\, q_i^T q_j/m\big)$ of the vector solutions, so $\operatorname{rank} X = \dim \operatorname{span}\{q_j\}$*, and the vector solutions of dimension one are the solutions, $q_j = x_j d$. Every vector solution gives a fractional exact cover $p_j = \|q_j\|^2 = d^T q_j$, with $Ap = \mathbf{1}$, $p \ge 0$ and $\sum_{j \in K} p_j \le 1$ for every clique K of Γ.*

Proof. Variables that share an equation contradict, so the vectors of one equation are pairwise orthogonal and $\sum_{j\in e}\|q_j\|^2 = \|d\|^2 = 1$.

Given a vector solution, let $v_j = z_j q_j/\sqrt{m}$ and let X be their Gram matrix. It is positive semidefinite and vanishes on the contradictions, i.e. off E, and $\operatorname{tr} X = \frac{1}{m}\sum_j w_j\|q_j\|^2 = \frac{1}{m}\sum_{e\in\mathcal{E}}\sum_{j\in e}\|q_j\|^2 = 1$. Moreover $\sum_j z_j v_j = \frac{1}{\sqrt{m}}\sum_j w_j q_j = \frac{1}{\sqrt{m}}\sum_{e\in\mathcal{E}}\sum_{j\in e} q_j = \sqrt{m}\,d$, so $\langle zz^T, X\rangle = \|\sum_j z_j v_j\|^2 = m$. Hence $\vartheta(\bar{G}, w) \geq m$, and $\vartheta(\bar{G}, w) = m$ by Proposition 6.11(b); X is a Lovász optimum.

Conversely, let $\vartheta(\bar{G}, w) = m$ and let X be a Lovász optimum. H_A is Lovász-optimal (Proposition 6.11(c)), so by (4) the range of X lies in L_A. Factor $X = VV^T$ with V of full column rank and let v_j be the rows of V. The columns of V lie in the range of X, so every column of $D^{-1/2}V$ is a u with $Au \in \mathbb{R}\mathbf{1}$, and the vectors v_j/z_j have one and the same sum t over every equation. Then $\sum_j z_j v_j = \sum_j w_j\, v_j/z_j = \sum_{e\in\mathcal{E}}\sum_{j\in e} v_j/z_j = mt$, and $m = \langle zz^T, X\rangle = \|mt\|^2$ gives $\|t\| = 1/\sqrt{m}$. So $q_j = \sqrt{m}\, v_j/z_j$ and $d = \sqrt{m}\, t$ form a vector solution: $q_i^T q_j = mX_{ij}/(z_i z_j) = 0$ on the contradictions. The two constructions are inverse to each other up to an orthogonal transformation of $\mathbb{R}^r$.

If the q_j span a line, $q_j = \lambda_j d$, with $\lambda_i\lambda_j = 0$ on the contradictions and $\sum_{j\in e}\lambda_j = 1$; at most one λ_j per equation is nonzero, so $\lambda \in \{0,1\}^n$ is a solution. Finally, for j in an equation e, $d^T q_j = \sum_{i\in e} q_i^T q_j = \|q_j\|^2$, so $p_j = \|q_j\|^2 = d^T q_j \geq 0$ does not depend on e and sums to $\|d\|^2 = 1$ over every equation. On a clique K of Γ the nonzero q_j are pairwise orthogonal, and Bessel's inequality gives $\sum_{j\in K} p_j = \sum_{j\in K,\, q_j\neq 0}(d^T q_j/\|q_j\|)^2 \leq \|d\|^2 = 1$. □

A vector solution is the semidefinite relaxation of the 0-1 system: each x_j becomes a vector q_j, $x_i x_j = 0$ becomes orthogonality and $\sum_{j\in e} x_j = 1$ becomes $\sum_{j\in e} q_j = d$. So the gap $\alpha < \vartheta = m$ of [Bu07] is the existence of vector solutions without solutions. By the last statement it implies that the fractional exact cover polytope $\{p \geq 0 :\ Ap = \mathbf{1},\ p(K) \leq 1 \text{ for all cliques } K \text{ of } \Gamma\}$ is nonempty, i.e. $\kappa(\Gamma, w) = m$ in the notation of [Bu07].

SAT01 propagation [Bu07, Section 3] reduces instances by unit propagation and clique analysis and adds edges to Γ by contradiction analysis: j and k contradict if some equation consists of variables that contradict j or k. Every solution survives, so α is unchanged, but a vector solution need not have $q_j^T q_k = 0$ for such a derived pair, so the new edges can destroy all vector solutions and push ϑ below m. Conversely, propagation can stop short of refuting an instance with $\vartheta < m$. Section 7.7 shows that on factoring instances the two refute different primes, and that their combination refutes primes that neither refutes alone.

6.6 Where the hardness lies

Propositions 6.10 and 6.11 raise a question. A Lovász-optimal wrapper is computable in polynomial time, to any precision, by semidefinite programming. If it is proper, [CW, Propositions 1 and 2] make the nonnegative points of its sphere of global minimizers clique indicators, and a SAT01 instance with $\vartheta = m$ is then satisfiable iff that sphere has a nonnegative point. So either deciding whether such a sphere has a nonnegative point is hard, or the Lovász-optimal wrappers of SAT01 instances are in general not proper. Both hold, but only the first is essential: it holds already where the Lovász-optimal wrapper is proper and needs no semidefinite program.

Proposition 6.13. *Let $H \in \mathcal{H}(G, w)$ be Lovász-optimal, $T = \ker(\vartheta(\bar{G}, w)I - H)$, and*

$$\Sigma_H = \{x \in T :\ z^T x = 1,\ x^T x = 1/\vartheta(\bar{G}, w)\}, \qquad P_H = \{x \in T :\ z^T x = 1,\ x \geq 0\}.$$

(a) Σ_H *is the set of global minimizers of (5), a sphere (possibly a single point) in the affine space* $\{x \in T : \ z^T x = 1\}$*, centred at its point of least norm.*

(b) *If the defects of* H *are nonnegative,* $x^T x \le 1/\vartheta(\bar{G}, w)$ *on the polytope* P_H*, so the nonnegative points of* Σ_H *are the points of* P_H *of largest norm, and they are vertices of* P_H*.*

(c) *If* H *is proper, the nonnegative points of* Σ_H *are exactly the indicators* x^Q *of the cliques with* $W(Q) = \vartheta(\bar{G}, w)$*; in particular* Σ_H *has a nonnegative point iff* $\omega(G, w) = \vartheta(\bar{G}, w)$*.*

Proof. (a) By Theorem 6.1, $v(H) = 1/\vartheta(\bar{G}, w)$, and a point of $\mathcal{W}^H$ has $1 = x^T H x \le \vartheta(\bar{G}, w)\, x^T x$, with equality iff $x \in T$. Conversely a point of T with $z^T x = 1$ and $x^T x = 1/\vartheta(\bar{G}, w)$ has $x^T H x = \vartheta(\bar{G}, w)\, x^T x = 1$. Within the affine space $x^T x$ is the squared distance from the origin, so the points of norm $1/\sqrt{\vartheta(\bar{G}, w)}$ form a sphere about its point of least norm.

(b) P_H lies in the bounded set $\{x \ge 0 : \ z^T x = 1\}$. For x there, $x^T H x \le 1$ by [CW, Proposition 1], and on T, $x^T H x = \vartheta(\bar{G}, w)\, x^T x$. A strictly convex function attains its maximum over a polytope only at vertices.

(c) Let $x \in \Sigma_H$ be nonnegative. Then $x^T H x = 1$, so its support is a clique Q by [CW, Proposition 2], and by the Cauchy–Schwarz inequality

$$1 = z^T x = \sum_{i \in Q} z_i x_i \le W(Q)^{1/2} (x^T x)^{1/2} = \big(W(Q)/\vartheta(\bar{G}, w)\big)^{1/2} \le \big(\omega(G, w)/\vartheta(\bar{G}, w)\big)^{1/2} \le 1.$$

Equality throughout gives $W(Q) = \omega(G, w) = \vartheta(\bar{G}, w)$ and x proportional to z_Q, i.e. $x = x^Q$. Conversely, if $W(Q) = \vartheta(\bar{G}, w)$ then x^Q is a global minimizer (Theorem 6.3), and it is nonnegative. □

So for a proper Lovász-optimal wrapper, $\omega(G, w) = \vartheta(\bar{G}, w)$ is decided by maximizing the norm over the polytope P_H. Everything convex is computed in polynomial time: $\vartheta(\bar{G}, w)$, H and T by the semidefinite program, P_H by linear algebra. What remains is to find a vertex of P_H of largest norm, the maximization of a convex function over a polytope, which is NP-hard in general. In terms of Corollary 6.4 and Theorem 6.12 it is the step from a Lovász optimum to a rank-one Lovász optimum. That this step carries the hardness, and not a lack of properness, is shown by the instances whose contradictions all lie in equations:

Corollary 6.14. *For every SAT01 instance, the nonnegative points of the sphere* $\Sigma_A = \{x \in L_A : \ z^T x = 1, \ x^T x = 1/m\}$*, which linear algebra computes from* A*, are exactly the indicators* x^S *of the exact covers* S *of* A*. Deciding whether* Σ_A *has a nonnegative point is therefore the exact cover problem, which is NP-complete [Ka72]. If every contradiction lies in an equation,* H_A *is proper, and it is Lovász-optimal with* $\Sigma_{H_A} = \Sigma_A$ *whenever* $\vartheta = m$*, in particular on every satisfiable instance.*

Proof. The last part of the proof of Proposition 6.11(d) uses only $x \in L_A$, $z^T x = 1$, $x^T x = 1/m$ and $x \ge 0$. By Proposition 6.11(a) the defect $mM_{ij}/(w_i w_j)$ of a contradiction is positive iff it shares an equation, and by (b) and (c) H_A is Lovász-optimal with $\ker(mI - H_A) = L_A$ when $\vartheta = m$. □

For QCP this is Proposition 6.10 and the remark after it; the corollary carries it over to every exact cover problem. (Whether ϑ equals m exactly is a question of precision that the corollary does not need: on the satisfiable instances it does.) With contradictions outside the equations properness fails as well, and the optimal face can show it:

Proposition 6.15. *Let X be a Lovász optimum with range R. If the sphere $\{x \in R : z^T x = 1,\ x^T x = 1/\vartheta(\bar{G}, w)\}$ has a nonnegative point whose support is not a clique of G, no Lovász-optimal wrapping matrix is proper, and every Lovász-optimal one with nonnegative defects has defect 0 on every non-adjacent pair inside that support.*

Proof. By (4), $R \subseteq \ker(\vartheta(\bar{G}, w)I - H)$ for every Lovász-optimal H, so such a point x has $x^T H x = \vartheta(\bar{G}, w)\, x^T x = 1$. With $z^T x = 1$,

$$x^T H x = (z^T x)^2 - \sum \tau_{ij} z_i z_j x_i x_j = 1 - \sum \tau_{ij} z_i z_j x_i x_j,$$

the sums running over the ordered non-adjacent pairs $i \neq j$, so the last sum vanishes. If the defects are nonnegative, so are its terms, and they all vanish; a proper H would then make the support of x a clique. □

The Lovász optima of Section 7.7 supply such points, numerically. On propagated Hamilton cycle instances, alternating projections between the sphere of Proposition 6.15 and the nonnegative orthant reached points whose negative part was below 10^{-4} of their norm and that are not solutions: 18 such points on the cubic graph with 20 vertices, of which at most 2 are solutions, and 4 on the dodecahedron, none of them a solution (`bench/jam.py`, Section 4 of `bench/runs/jam/jam-2026-10-03.txt`). So these instances have no proper Lovász-optimal wrapper. Properness fails on the contradictions outside the equations, to which H_A is blind (Proposition 6.11(a)), but by Corollary 6.14 this is not what keeps SAT01 hard.

Searching for nonnegative points of the sphere is thus a heuristic for an NP-hard problem. QUALEX-MS's Douglas–Rachford stage (`QMS_DR`) drives the corners of each degenerate sphere of stationary points towards the nonnegative orthant, and on H_A it can use instead a greedy projection that also respects the contradictions outside the equations (`sat01qms -D`). On the propagated satisfiable instances where the greedy stage of QUALEX-MS fails, it found solutions of the factoring instances 14471, 32399 and 60491, whose degenerate top clusters have dimension 148 to 235, and of the cubic graph with 30 vertices (dimension 580), none of which QUALEX-MS solves without it. It reached 911 of $m = 912$ on $1000001 = 101 \cdot 9901$ (dimension 471) and found nothing on the cubic graphs with 40 and 50 vertices (dimensions 1178 and 1828) (`bench/sat01.py --qms`, `bench/runs/sat01/qms.tsv`).

6.7 A second wrapper surface

A solution of a SAT01 instance lies not only on Σ_A but, being a clique indicator, on the surface $\mathcal{W}^H$ of every $H \in \mathcal{H}(G, w)$ (Section 2). A proper H sees the contradictions that H_A does not:

Proposition 6.16. *For every SAT01 instance and every proper $H \in \mathcal{H}(G, w)$, the nonnegative points of $\Sigma_A \cap \mathcal{W}^H$ are exactly the indicators x^S of the solutions S.*

Proof. By Corollary 6.14 the nonnegative points of Σ_A are the x^S of the exact covers S. For $x \geq 0$ with $z^T x = 1$, $x^T H x = 1 - \sum \tau_{ij} z_i z_j x_i x_j$ over the ordered non-adjacent pairs, as in the proof of Proposition 6.15, and with all defects positive this is 1 iff the support of x contains no non-adjacent pair. So $x^S \in \mathcal{W}^H$ iff S contains no contradiction, i.e. iff S is a solution. □

With the standard wrapper H_0, therefore, the sphere, the orthant and $\mathcal{W}^{H_0}$ meet exactly in the solutions, while the sphere and the orthant alone meet in the exact covers. (When every contradiction lies in an equation, every exact cover is a solution and the surface removes nothing.) Projecting onto a wrapper surface is the relaxed program with its origin moved:

Lemma 6.17. *Let $H \in \mathcal{H}(G,w)$ and $y \in \mathbb{R}^n$, let $\hat H = U\mathrm{diag}(\lambda_i)U^T$ and $c = U^T\hat b$ as in Section 2, $\beta = U^TPy$, and let J be the open interval of the t with $1+t\lambda_i > 0$ for every i, on which $I+t\hat H$ is positive definite on $z^\perp$ (between $-1/\lambda_{\max}(\hat H)$ and $-1/\lambda_{\min}(\hat H)$ when $\lambda_{\max}(\hat H) > 0 > \lambda_{\min}(\hat H)$). For $t \in J$ let*

$$u_i(t) = \frac{\beta_i - tc_i}{1+t\lambda_i}, \qquad \phi(t) = \sum_i (\lambda_i u_i + 2c_i)\, u_i - s.$$

Then $\phi'(t) = -2\sum_i(\lambda_i u_i + c_i)^2/(1+t\lambda_i) \le 0$, and if $\phi(t^*) = 0$, the point $x^0 + Uu(t^*)$ is a point of $\mathcal{W}^H$ nearest to y.

Proof. For $x = x^0 + \hat x$ with $\hat x \perp z$, $\|x-y\|^2 = \|\hat x - Py\|^2 + \|x^0 - (I-P)y\|^2$, and the second term does not depend on x. For $t \in J$ the function $L(\hat x) = \|\hat x - Py\|^2 + t(q(\hat x) - s)$ is convex on $z^\perp$, with Hessian $2(I+t\hat H)$, and its minimizer solves $(I+t\hat H)\hat x = Py - t\hat b$, i.e. $\hat x = Uu(t)$, where $q(\hat x) - s = \phi(t)$. If $\phi(t^*) = 0$, every $\hat x'$ with $q(\hat x') = s$ has $\|\hat x' - Py\|^2 = L(\hat x') \ge L(Uu(t^*)) = \|Uu(t^*) - Py\|^2$. The derivative follows from $u_i' = -(\lambda_i u_i + c_i)/(1+t\lambda_i)$. □

At the ends of J, ϕ tends to $+\infty$ and $-\infty$ unless $\beta_i - tc_i$ vanishes there on the extreme eigenvectors, so in general it has exactly one root. QUALEX-MS computes it with one eigendecomposition of $\hat H$ and a safeguarded Newton iteration per point (`wrapper_surface()` in `lib/wrapper.cc`), and its Douglas–Rachford stage takes such surfaces as further sets: each start keeps one copy on the sphere and one on the surface, and the orthant or the 2-clause projection acts on the average of their reflections, which is Douglas–Rachford between the product of the sphere and the surface and the diagonal of the third set (without the surface, the plain iteration). With $H = H_0$ (`sat01qms -W`), on the instances of the previous subsection, the orthant now finds the solution of 14471 and reaches 513 of $m = 514$ on 60491 and 905 of 912 on 1000001, against 364, 505 and 895 without the surface; with the 2-clause projection it solves the same seven instances as before. The surface without nonnegativity (`sat01qms -N`) finds nothing that the run without a Douglas–Rachford stage does not, and the iteration converges poorly between the two quadrics: on the factoring instances 1 to 19 of the 296 to 942 starts meet the stopping test (10^{-9}) within 300 iterations, and on the Hamilton cycle instances all of them do, after 123 to 190 iterations on average (QUALEX-MS commits 892a830 and 02a809e, `bench/runs/sat01/qms.tsv`). Douglas–Rachford is the wrong tool for that intersection, which is explicit:

Proposition 6.18. *Let $\Sigma = \{\bar x + Qp :\ p \in \mathbb{R}^k,\ \|p\| = \rho\}$ with $Q^TQ = I$ and $Q^T\bar x = 0$, let H be symmetric, and write, for $x = \bar x + Qp$,*

$$x^THx - 1 = p^TMp + 2g^Tp + h, \qquad M = Q^THQ,\quad g = Q^TH\bar x,\quad h = \bar x^TH\bar x - 1.$$

Let $M = V\mathrm{diag}(\kappa_i)V^T$, $\gamma = V^Tg$, and let σ, not an eigenvalue of M, solve

$$\sum_i \frac{\gamma_i^2}{\kappa_i - \sigma} = h + \sigma\rho^2. \tag{8}$$

Let $v_i = \gamma_i/(\sigma - \kappa_i)$ and $I_\pm = \{i :\ \pm(\kappa_i - \sigma) > 0\}$, and for unit vectors $\xi \in \mathbb{R}^{I_+}$ and $\eta \in \mathbb{R}^{I_-}$ let $w_i = \xi_i/\sqrt{\kappa_i - \sigma}$ on I_+ and $w_i = \eta_i/\sqrt{\sigma - \kappa_i}$ on I_-. Then the points of Σ with $x^THx = 1$ are the points $\bar x + QV(v + \nu w)$ for all such ξ, η and all real roots ν of

$$\|w\|^2\nu^2 + 2\,v^Tw\,\nu + \|v\|^2 - \rho^2 = 0,$$

together with $\bar{x} + QVv$ *if* $\|v\| = \rho$. *If* $\|v\| < \rho$, *each of these equations has exactly one positive root* ν_+, *and* $(\xi, \eta) \mapsto \bar{x} + QV(v + \nu_+ w)$ *maps the product of the unit spheres of* $\mathbb{R}^{I_+}$ *and* $\mathbb{R}^{I_-}$ *one-to-one onto* $\Sigma \cap \{x^T H x = 1\}$. *Equation (8) has a root between any two consecutive distinct eigenvalues of* M *on whose eigenspaces* g *does not vanish.*

Proof. On Σ, $\|p\|^2 = \rho^2$, so for $u = V^T p$, $K = \mathrm{diag}(\kappa_i)$ and every σ,

$$x^T H x - 1 = u^T (K - \sigma I) u + 2\gamma^T u + h + \sigma\rho^2 = (u - v)^T (K - \sigma I)(u - v) - v^T (K - \sigma I) v + h + \sigma\rho^2,$$

since $(K - \sigma I)v = -\gamma$. As $v^T (K - \sigma I) v = \sum_i \gamma_i^2 / (\kappa_i - \sigma)$, (8) makes the last two terms cancel, and on Σ

$$x^T H x - 1 = \sum_{i \in I_+} (\kappa_i - \sigma)(u - v)_i^2 - \sum_{i \in I_-} (\sigma - \kappa_i)(u - v)_i^2.$$

This vanishes at $u = v$; for $u \neq v$ the two sums are not both zero, since every index lies in $I_+ \cup I_-$, so it vanishes iff they are equal to some $\nu^2 > 0$, i.e. iff $u - v = \nu w$ with $\xi_i = \sqrt{\kappa_i - \sigma}\,(u - v)_i / \nu$ and $\eta_i = \sqrt{\sigma - \kappa_i}\,(u - v)_i / \nu$ unit vectors. Conversely each such w makes both sums equal to 1, so $v + \nu w$ lies on the cone for every ν, and it lies on Σ iff $\|v + \nu w\| = \rho$. If $\|v\| < \rho$, the constant term of that quadratic is negative and its roots have opposite signs. Distinct (ξ, η) give distinct rays from v, because the normalization fixes w within its ray, and every point u of the intersection lies on the ray from v through it ($u \neq v$, as $\|v\| < \rho = \|u\|$), so the map is one-to-one and onto. Finally the left side of (8) minus its right side is continuous between two consecutive distinct eigenvalues $\kappa_a < \kappa_b$, and tends to $-\infty$ at κ_a and to $+\infty$ at κ_b when g has a component in both eigenspaces. □

Proposition 6.18 is Levin's pencil method for the intersection of two quadrics [Le76], carried to dimension k. Every member $x^T H x - 1 - \sigma(\|p\|^2 - \rho^2)$ of the pencil spanned by the two quadrics cuts Σ in the same set. (8) picks a singular member, a cone, and each line of the cone through its vertex meets Σ in the roots of one quadratic. For quadrics in projective 3-space, Dupont, Lazard, Lazard and Petitjean made the method exact and near-optimal in the number of radicals [DLLP08]. In the generic case they parametrize a member of inertia $(2, 2)$, whose two families of lines give a bilinear parametrization, rather than a cone, whose vertex needs a root of the pencil's determinantal equation, and they classify the singular intersections. Lazard, Peñaranda and Petitjean implemented it as QI [LPP06]. In floating point and in dimension k neither refinement matters, and the cone is the simplest member to parametrize.

The intersection thus has dimension $|I_+| + |I_-| - 2 = k - 2$, and for $\Sigma = \Sigma_A$ (with $\bar{x}$ its centre, Q an orthonormal basis of $\{x \in L_A : z^T x = 0\}$ and $\rho^2 = 1/m - \|\bar{x}\|^2$) and $H = H_0$ it costs one eigendecomposition of order k and one scalar equation. `bench/cone.py` checks it (`bench/runs/sat01/cone-2026-10-10.txt`). On 14471, where $k = 148$, (8) has a root in 145 of the 147 gaps of the spectrum of M. On the five instances checked (14471, 32399, 60491, the dodecahedron and $GP(13, 2)$), the points of Proposition 6.18 satisfy both equations to within $2 \cdot 10^{-13}$, and the solution found by exact cover search lies on the cone with its own (ξ, η). But on all five every root puts the vertex v outside the ball, at 2.8ρ to 7.4ρ for the nearest, so the generators that reach the sphere form a thin set: none of 10^5 random ones did, while those near a solution's do. And leaving the solution along the intersection brings negative entries at once. Perturbing its (ξ, η) by 10^{-3} puts a median 3 to 5% of the ℓ_1 mass of the point on negative entries on the factoring instances, and 16 to 19% on the Hamilton cycle ones.

So the two quadratic conditions cost an eigendecomposition and a root of (8), and by Proposition 6.16 the solutions are exactly the parameters (ξ, η) whose point is nonnegative. All of the hardness is in that last condition: when every contradiction lies in an equation the surface removes no exact cover, and deciding whether the intersection has a nonnegative point is the exact cover problem (Corollary 6.14).

6.8 The smallest jam

By (4) the range of every Lovász optimum lies in the top eigenspace of every Lovász-optimal wrapping matrix. No such matrix, therefore, has a top eigenspace smaller than the subspace R^* spanned by the ranges of all Lovász optima, and R^* contains every solution. It is the range of an optimum of maximal rank, to which the central path of (2)–(3) converges [HKR02], so an interior point method such as CSDP approaches it. When $\vartheta = m$, H_A already has the optimal eigenvalue, and the problem shrinks.

Proposition 6.19. *Let $\vartheta = m$, let the columns of B be an orthonormal basis of L_A, $d = \dim L_A$, $\mathcal{S}^d$ the symmetric $d \times d$ matrices, b_i the rows of B, $S_{ij} = (b_i b_j^T + b_j b_i^T)/2$, and*

$$V = \{Y \in \mathcal{S}^d : \ \langle S_{ij}, Y\rangle = 0 \text{ for every contradiction } \{i,j\}\},$$
$$F = \{Y \in V : \ Y \succeq 0, \ \operatorname{tr} Y = 1\}.$$

(a) The Lovász optima are exactly the matrices BYB^T with $Y \in F$.

(b) If $Y^* \in F$ has maximal rank, $R^* = B \operatorname{range}(Y^*)$, and it contains the indicator x^S of every solution S.

(c) If $W \succeq 0$ is orthogonal to V, then $WY = 0$ for every $Y \in F$. If moreover $\operatorname{rank} Y + \operatorname{rank} W = d$ for some $Y \in F$, then $R^* = B \operatorname{range}(Y) = B \ker W$.

(d) The matrices orthogonal to V are the $W = B^T S B$ with S symmetric and supported on the contradictions, and then $H_A - tS \in \mathcal{H}(G, w)$ and $B^T(H_A - tS)B = mI - tW$ for every t.

Proof. (a) H_A is Lovász-optimal by Proposition 6.11(c), so by (4) every optimum X has its range in L_A: $X = BYB^T$ with $Y = B^T X B \succeq 0$ and $\operatorname{tr} Y = \operatorname{tr} X = 1$, and $X_{ij} = b_i^T Y b_j = \langle S_{ij}, Y\rangle$ vanishes on the contradictions. Conversely such an X is feasible in (3), and as $H_A - zz^T$ vanishes on the diagonal and off the contradictions, $\langle zz^T, X\rangle = \langle H_A, X\rangle = m \operatorname{tr} X = m$.

(b) F is convex, and the range of $(Y_1 + Y_2)/2$ is the sum of the ranges of $Y_1, Y_2 \in F$, so a point of maximal rank has a range that contains the range of every other. By Lemma 2.2, $z_S z_S^T/m$ is an optimum, and $z_S = m\, x^S$.

(c) $\langle W, Y'\rangle = 0$ and $W, Y' \succeq 0$ give $WY' = 0$, so every $Y' \in F$ has its range in $\ker W$, and $R^* \subseteq B \ker W$. As $Y \in F$ has rank $d - \operatorname{rank} W = \dim \ker W$, $B \operatorname{range}(Y) \subseteq R^*$ is all of $B \ker W$.

(d) $V^\perp$ is spanned by the $S_{ij} = B^T(e_i e_j^T + e_j e_i^T)B/2$ of the contradictions, which are nonedges of G, where the entries of a wrapping matrix are free; and $B^T(mI - H_A)B = 0$ by Proposition 6.11(b). □

So R^* is the most that minimizing the jam of H_A can reach. A certificate W of (c) is a step of facial reduction [BW81] for F, and by (d) it is the first-order part of a perturbation of H_A in its free entries, which keeps the top eigenvalue m on $B \ker W$ and lowers it on $B \operatorname{range}(W)$, up to

the coupling of L_A with $L_A^\perp$, which is of order t^2. Where CSDP resolves the optimal wrappers, on the Hamilton cycle instances of Table 8, one of them has R^* as its top eigenspace.

F lies in $\mathcal{S}^d$ rather than in the symmetric $n \times n$ matrices, has no objective, and is cut by one equation per contradiction, most of them redundant. V can be computed exactly. With a basis N of $\{u : \ Au \in \mathbb{R}\mathbf{1}\}$ in place of B the constraints read $n_i^T Y n_j = 0$, the n_i being the rows of N, since the factor $D^{1/2}$ drops out. Gaussian elimination of A with a Markowitz pivot rule gives a rational N whose rows have 4 to 32 nonzeros on average, and sparse elimination of the constraint rows modulo the prime $2^{61} - 1$, again with a Markowitz rule, gives their rank, hence $\dim V$, together with a set Γ' of contradictions whose constraints are independent and imply all the others. (Independence modulo a prime implies independence over the rationals, and the rank modulo a prime is smaller only if the prime divides all the nonzero maximal minors.) Then R^* and a certificate come from the pair

$$\begin{aligned} &\max\ t \quad \text{s.t.} \quad Y - tI \succeq 0, \quad \langle S_{ij}, Y\rangle = 0 \text{ for } \{i,j\} \in \Gamma', \quad \operatorname{tr} Y = 1, \\ &\min\ y_0 \quad \text{s.t.} \quad W = \sum_{\{i,j\}\in\Gamma'} y_{ij} S_{ij} + y_0 I \succeq 0, \quad \operatorname{tr} W = 1. \end{aligned}$$

Both are strictly feasible ($W = I/d$ in the second). Unless some Lovász optimum is positive definite on L_A their common value is 0, the optimal Y form F, the optimal W are the certificates of (c) with trace 1, and a primal-dual interior point method converges to a pair of maximal ranks. Its Schur matrix has order $|\Gamma'| + 1$, against one row per contradiction for (3), and as the S_{ij} have rank two its entries are products of entries of BZB^T and $BW^{-1}B^T$ ($Z = Y - tI$), as in CSDP. We used the HKM direction [HRVW96] with Mehrotra's predictor-corrector [Me92].

Table 9 in Section 7.7 gives the results. On every instance where the method finished, $\operatorname{rank} Y + \operatorname{rank} W = d$, so R^* is proved minimal to the accuracy of the solver. On 14471 and $179 \cdot 181$, R^* is the span of the solutions, and F is a point or the segment between the two solutions; on the cubic graph with 16 vertices V itself is spanned by the 12 solutions and F is their simplex. There the semidefinite program solves the instance. Everywhere else R^* keeps 91 to 95% of L_A: 214 of its 236 dimensions on $241 \cdot 251$, 445 of 472 on 1000001, and all but 8 to 24 on the Hamilton cycle instances.

Those are the instances on which the search of Section 6.6 needs help, and the smaller sphere $\{x \in R^* : \ z^T x = 1, \ x^T x = 1/m\}$, which holds every solution, gives it little. Douglas–Rachford with the 2-clause projection from the $2(\dim R^* - 1)$ corners of that sphere (the iteration of Section 6.6 in numpy, `bench/minjam.py dr`: 300 iterations, MIN at the end) found a solution of $241 \cdot 251$ from 28 of 426 corners, against 18 of 470 on Σ_A, and none of 1000001 either way (911 of 912 at best). On the Hamilton cycle instances the count moved both ways: from 190 to 138 on the cubic graph with 20 vertices, 91 to 76 on the dodecahedron, 28 to 34 on the cubic graph with 24 vertices and 1 to 0 on $GP(13, 2)$. With the orthant at most 6 corners succeeded on either sphere. (The corners depend on the orthonormal basis chosen in each sphere, and with another basis these counts moved by up to 15.)

A first-order method is no shortcut to R^*. The factorization $Y = CC^T/\|C\|^2$ of [BM03] (`bench/minjam.py bm`), minimizing $\sum_{\{i,j\}} \langle S_{ij}, Y\rangle^2$ over $C \in \mathbb{R}^{d\times d}$ by L-BFGS at $O(n^2 d)$ per step, reaches F quickly: MIN on the top eigenvectors of Y finds the solution of 14471 after 7 seconds and a solution of $179 \cdot 181$ after 48, against CSDP's 306 and 659, and it reaches the face of 1000001. But it converges to the relative boundary of F, where Y loses rank (on $241 \cdot 251$ its limit misses the solutions by 3%), and only to about 10^{-7}, which hides the directions of

R^* whose eigenvalues are near 10^{-6}. Averages of several limits have full rank, and neither the residual nor a variant with $Y \succeq \delta I$ yields a certificate.

So the smallest jam is computable, exactly and with a certificate, also where CSDP is not, but it is small only where the instance is easy anyway. Minimizing the jam leaves the step from the optimal face to a rank-one point (Section 6.6) where it was.

7 Computations

7.1 Tools

The maximum cliques were listed with an unpublished maximal clique enumerator (CLEN). The system (7) with a common level was set up with repeated equations removed and tested for a solution at level 1 (least squares, confirmed by linear programming); when solvable, the spectrum was examined for the solution nearest the standard wrapper (least change of the defects, level free), and nonnegative solutions were explored by linear programming (largest smallest defect, and the defects that vanish in every nonnegative solution). ϑ was computed with CSDP [Bo99] from whichever of two equivalent programs has fewer constraints: (3), one constraint per non-edge, or $\min t$ subject to $tI - H \succeq 0$, $H \in \mathcal{H}(G, w)$, i.e. a positive semidefinite $Y = tI - H$ with $y_{ii} = t-1$ and $y_{ij} = -1$ on E, one constraint per edge. The Lovász-optimal H was recovered from the solution. The scripts are `bench/anchor_all.py` and `bench/theta.py` of the QUALEX-MS repository (commit c9af3f8), and their outputs are `bench/runs/anchor-all-2026-09-26.txt` and `bench/runs/theta/theta.tsv`.

The following checks agree with the statements above: Theorem 5.1 to $4 \cdot 10^{-13}$ on 127 pairs of maximal cliques of weighted $G(40, 0.5)$ that admit nonzero levels; the criterion of Theorem 5.4 on all 150 pairs tried (23 of which admit only level 0); Proposition 3.5 to 10^{-15} on 1997 pairs of cliques; Theorem 5.3 with multiplicity equal to d and $|c| \le 10^{-16}$ on the eigenspace; Theorem 6.6 and the even hole statement for $C_5, \dots, C_{10}$ and $\bar{C}_7, \bar{C}_9, \bar{C}_{11}$; for C_5, over a grid of the free parameters of one anchored edge, $\lambda_{\max}(\hat{H}) - \mu \ge 0.138$ (Theorem 6.3, $\vartheta = \sqrt{5} > 2$); the colouring wrapper on random interval graphs; and Theorem 6.1 and Corollary 6.2 on the graphs of Table 5. CSDP was validated on `hamming6-2` (32), `johnson8-4-4` (14), `hamming6-4` (16/3) and `MANN_a9` (17.47503), with both programs and with CSDP's own `theta` program.

7.2 Anchoring all maximum cliques of DIMACS graphs

Table 3 covers every DIMACS graph with $n \le 1100$ and between 2 and 20000 maximum cliques. For 11 of them the common level is forced to 0; by Corollary 6.8 all have $\vartheta > \omega$. For the other 20 the level is free. On 6 of these the anchoring nearest the standard wrapper already puts μ at the top of the spectrum of $\hat{H}$, which by Theorem 6.3 certifies $\vartheta = \omega$; for five of them the standard wrapper itself anchors all maximum cliques, because every outside vertex has the same number of non-neighbours in each. (For `johnson8-4-4` the maximum cliques are the 30 Steiner systems $S(3, 4, 8)$, and a 4-set outside such a system meets exactly 4 of its 14 blocks in three points.) On the remaining 14 the nearest anchoring leaves between 2 and 232 eigenvalues above μ, and CSDP decides: $\vartheta = \omega$ on 8 of them and $\vartheta > \omega$ on 6. Wherever the level is free, the multiplicity of μ in $\hat{H}$ equals d exactly, as Theorem 5.3 predicts as a lower bound, with the single exception of `hamming8-4` (127 against 71).

Two features of the solutions are worth recording. First, Proposition 5.6 is not an exception.

graph	n	ω	k	d	common level	above μ	$\vartheta(\bar{G})$
p_hat300-1	300	8	13	12	forced to 0		$> \omega$
sanr200_0.7	200	18	13	10	forced to 0		$> \omega$
p_hat300-2	300	25	52	17	forced to 0		$> \omega$
DSJC500.5	500	13	56	55	forced to 0		$> \omega$
p_hat500-1	500	9	78	74	forced to 0		$> \omega$
p_hat700-2	700	44	138	23	forced to 0		$> \omega$
hamming6-4	64	4	240	51	forced to 0		5.33333
p_hat1000-1	1000	10	276	260	forced to 0		$> \omega$
C125.9	125	34	924	24	forced to 0		$> \omega$
keller4	171	11	2304	133	forced to 0		$> \omega$
MANN_a9	45	16	9540	44	forced to 0		17.47503
hamming6-2	64	32	2	1	free, standard	0	32
hamming8-2	256	128	2	1	free, standard	0	$= \omega$
hamming10-2	1024	512	2	1	free, standard	0	$= \omega$
johnson8-4-4	70	14	30	14	free, standard	0	14
johnson8-2-4	28	4	105	20	free, standard	0	$= \omega$
hamming8-4	256	16	480	71	free	0	$= \omega$
san200_0.7_2	200	18	2	1	free	43	18
gen200_p0.9_55	200	55	4	2	free	13	55
gen200_p0.9_44	200	44	4	2	free	33	44
c-fat200-1	200	12	14	13	free	8	12
c-fat500-1	500	14	19	18	free	19	14
c-fat500-2	500	26	19	18	free	7	26
c-fat500-5	500	64	3	2	free	4	64
c-fat500-10	500	126	3	2	free	2	126
brock200_1	200	21	2	1	free	49	27.45664
p_hat700-1	700	11	2	1	free	232	15.11986
c-fat200-5	200	58	3	2	free	2	60.34528
sanr400_0.5	400	13	4	3	free	143	20.31766
p_hat300-3	300	36	10	5	free	90	41.16993
p_hat500-2	500	36	14	5	free	151	38.97466

Table 3: DIMACS graphs with $2 \leq k \leq 20000$ maximum cliques and $n \leq 1100$. d is the dimension of the affine hull of their indicators; "standard": the standard wrapper already anchors them all; "above μ": eigenvalues of $\hat{H}$ above the common level for the anchoring nearest the standard wrapper. ϑ: computed with CSDP where a value is given; $> \omega$ by Corollary 6.8; $= \omega$ by Theorem 6.3 (and, for the hamming graphs of distance 2, which are complements of bipartite graphs, by perfection).

Only 7 of the 20 graphs, the six symmetric ones and `brock200_1`, admit a proper anchoring of all their maximum cliques; on the other 13, between 2.6% (`san200_0.7_2`) and 23% (`c-fat200-5`) of the non-edges involved are invisible in every nonnegative solution. (The solutions (τ, μ) form a cone, so a single linear program finds these non-edges: maximize $\sum_k \min(\tau_k, 1)$ subject to (7), $\tau \geq 0$ and $\mu \geq 1$; exactly the defects that can be positive reach 1.) Second, in the graphs with the level forced to 0 the collapse is local and has small certificates, of the kind of Example 6.9.

7.3 Random graphs

On the 35 unweighted random graphs with proven ω ($n = 200$ with $p \leq 0.8$ and $n = 300$ with $p \leq 0.7$), 5 have a unique maximum clique. Of the 30 others the common level is forced to 0 on 17 (with k between 3 and 797) and free on 13 (with k between 2 and 15); where free, μ had 55 to 99 eigenvalues of $\hat{H}$ above it and the multiplicity of μ was d exactly. One of the 13 admitted no nonnegative solution, 5 admitted proper ones, and on the other 7 between 2.1% and 9.4% of the non-edges involved are invisible in every nonnegative solution. Weighted random graphs almost always have a unique maximum weight clique, for which anchoring is the single clique construction of Section 3.

7.4 The Lovász number of the 14 undecided graphs

Table 4 lists the CSDP results for the 14 graphs on which the nearest anchoring stayed below the top. In all eight graphs with $\vartheta = \omega$ the optimal wrapper anchors every maximum clique at $C = 0$ (to 10^{-9} to 10^{-4}), as Theorem 6.3 requires. The multiplicity of $\lambda_{\max}(H)$ agrees with the rank of the Lovász optimum found by CSDP (strict complementarity) and, but for one graph, with $\dim \operatorname{span}\{z_Q\}$, the lower bound of Corollary 6.4. The exception is `gen200_p0.9_44`: its Lovász optimum has rank 157 and a positive diagonal on all 200 vertices, so by (4) every Lovász-optimal wrapper has a top eigenvalue of multiplicity at least 157, though its four maximum cliques span only 3 dimensions. The optimal face of (3) contains "vector cliques" that are not cliques. When $\vartheta > \omega$ the top multiplicities are large and unrelated to the cliques; on the largest graphs the top of the spectrum decays gradually and the multiplicity is only defined up to the solver's accuracy.

7.5 Relaxed bounds of the standard and the optimal wrapper

Table 5 compares the bound $1/v(H)$ of Theorem 6.1 for the standard wrapper and for the optimal wrapper found by CSDP; $v(H)$ was computed from the secular equation as the stationary point with the largest multiplier (Lemma 2.3), including the hard case. For the optimal wrappers $1/v(H) = \vartheta$ to the accuracy of the solver (relative error below $3 \cdot 10^{-6}$); on the three regular graphs $1/v(H_0)$ equals Hoffman's bound, as Corollary 6.2 states; elsewhere the standard wrapper's bound can be very weak (`san200_0.7_2`: 115.8 against $\vartheta = 18$).

7.6 Quasigroups with holes

Proposition 6.10 places the least top multiplicity of a Lovász-optimal wrapper of a completable QCP graph between the span of the completions and $\dim L$. To see where it falls, we generated quasigroup-with-holes (QWH) instances [AGKS00]: a random latin square of order n from the Jacobson–Matthews chain [JM96], with h cells emptied uniformly at random, for $n = 10, 15, 20$ and hole fractions from 30 to 70% (four instances per cell, three for $n = 20$). The range covers

graph	ω	$\vartheta(G)$	$\dim \operatorname{span}\{z_Q\}$	rank of X^*	mult. of $\lambda_{\max}(H)$
`c-fat200-1`	12	12.000000	14	14	14
`c-fat500-1`	14	14.000000	19	19	19
`c-fat500-2`	26	26.000000	19	19	19
`c-fat500-5`	64	64.000001	3	3	3
`c-fat500-10`	126	126.000000	3	3	3
`gen200_p0.9_55`	55	55.000000	3	3	3
`gen200_p0.9_44`	44	44.000000	3	157	157
`san200_0.7_2`	18	18.000000	2	2	2
`brock200_1`	21	27.456641	2	63	63
`c-fat200-5`	58	60.345275	3	5	5
`p_hat300-3`	36	41.169930	6	101	101
`sanr400_0.5`	13	20.317656	4	≈ 200	≈ 200
`p_hat500-2`	36	38.974658	6	~ 150	~ 150
`p_hat700-1`	11	15.119858	2	~ 450	~ 450

Table 4: The Lovász number of the complement (CSDP, relative duality gaps below 10^{-8}), the dimension spanned by the maximum cliques, the rank of the Lovász optimum X^* and the multiplicity of the top eigenvalue of the optimal wrapper, both read at the largest relative gap in the spectrum. For `sanr400_0.5` that gap lies between 199 and 201. For the last two graphs there is no clear gap: counting eigenvalues within relative distances 10^{-6} to 10^{-4} gives 74 to 165 (`p_hat500-2`) and 428 to 486 (`p_hat700-1`). The largest problems had 61800 constraints and took 42 and 47 minutes.

the passage from a single completion to tens of thousands, where QWH instances are hardest for search [AGKS00]. The completions were listed by exact cover search; where there were more than 20000 of them, the span was measured on the first 20000 together with about 2000 sampled by randomized descents, restarted after a node limit because such searches have heavy-tailed run times. ϑ and a Lovász optimum were computed with CSDP. Table 6 summarizes the 68 instances; `bench/qwh.py` reproduces them.

Every instance confirms Proposition 6.10: $\vartheta(G) = h$, the standard wrapper in the gauge $J - \frac{h}{3}A_G$ has the top eigenvalue h with multiplicity exactly $\dim L$, and every completion is anchored at $C = 0$ to 10^{-13}.

The Lovász optimum spans exactly the completions whenever there are few of them: its rank equals their span on 29 of the 41 instances listed in full, among them all nine with a unique completion, where the optimum found by CSDP is the projector of that completion. It does so again when there are very many: on the 27 instances whose listing was capped, the listed and sampled completions span as much as the optimum on 26 of them and one dimension less on the other, and at 70% holes both fill all of L. In between, with tens to hundreds of completions, the optimum exceeds their span: 64 dimensions against 34, 40 against 18 and 75 against 60 on three instances with $n = 15$ and 45% holes, 15 against 7 on one with $n = 10$. Every Lovász-optimal wrapper of those graphs has top eigendirections that no completion spans. On a few of them complementarity is not strict: some directions carry both an eigenvalue of the optimum and a gap of the wrapper of order $\sqrt{\mu}$, and the rank is only bracketed (the parenthesized ranges of Table 6).

The natural wrappers are far more degenerate. The standard wrapper's top eigenspace L had dimension 111 on an instance whose two completions span 2 dimensions ($n = 20$, 35% holes), and

graph	ω	$\vartheta(G)$	$1/v(H_0)$	Hoffman
`hamming6-2`	32	32.000	32.000	32.000
`johnson8-4-4`	14	14.000	14.000	14.000
`hamming6-4`	4	5.333	13.538	13.538
`MANN_a9`	16	17.475	19.707	
`c-fat200-1`	12	12.000	17.430	
`gen200_p0.9_55`	55	55.000	75.624	
`gen200_p0.9_44`	44	44.000	72.547	
`san200_0.7_2`	18	18.000	115.759	
`brock200_1`	21	27.457	43.287	
`c-fat200-5`	58	60.345	72.905	
`p_hat300-3`	36	41.170	88.374	

Table 5: The relaxed bound $1/v(H)$ for the standard wrapper against ϑ, which the optimal wrapper attains (Theorem 6.1), and Hoffman's bound for the regular graphs.

194 against a span of 4 on another ($n = 20$, 40% holes); the colouring wrapper of the cells had 327 and 453. So on such instances only a handful of the degenerate directions of the standard QUALEX-MS wrapper carry information on the completions, as [BP03] anticipates. The least degenerate Lovász-optimal wrapper can be much sharper, but it too exceeds the completions in the middle of the transition. These instances are small: all nine with a unique completion were solved by propagation alone, so they say nothing about hard instances with a unique completion.

7.7 SAT01 instances

Two reductions of [Bu07] supplied the instances, built with Busygin's converters. A factoring instance encodes the long multiplication of two factors with product N, with the bitwise sums of [Bu07, Section 4.9]; trivial factorizations are excluded, so a prime gives an unsatisfiable instance, and the products of two primes tried had one solution, or two when both factors have the same bit length. A Hamilton cycle instance of a graph with n vertices has the variables x_{ij} ("the i-th vertex of the cycle is vertex j"), $2n$ equations that make (x_{ij}) a permutation matrix, and a contradiction between x_{ij} and $x_{i+1,k}$ whenever j and k are not adjacent [Bu07, Section 4.2]; one variable is fixed to remove the rotations, and once it is eliminated every weight is 2 and $m = 2(n-1)$. The graphs were generalized Petersen graphs $GP(n, k)$, among them the Petersen graph $GP(5, 2)$ and the dodecahedron $GP(10, 2)$, the flower snarks J_5 and J_7, the Coxeter graph, and random cubic graphs; the solver confirmed that the Petersen graph, $GP(11, 2)$, J_5, J_7 and the Coxeter graph have no Hamilton cycle.

Each instance was converted after SAT01 propagation of two strengths: *light*, which only eliminates the variables that the encoding fixes, and *full*, unit propagation, clique analysis and contradiction analysis to a fixed point, as in Busygin's solver. Full propagation refutes all 1728 primes between 2^{10} and 2^{14} but 14887, four of the eight largest primes below 2^{15}, and none of the eight largest below 2^{16} or below 2^{17}. It solves the 2957 odd products of two primes in the same range or reduces them to one equation in two variables, except 14471, 14631, 15591 and 16041. Bipartite graphs with sides of different sizes, which have no Hamilton cycle by parity, are refuted already by light propagation, whose clique analysis carries the parity of the positions from the fixed vertex.

n	holes	inst.	$\lvert V\rvert$	completions	span	rank of X^*	dim L	$\lvert V\rvert - h + 1$
10	30%	4	50–59	1–2	1–2	1–2	1–3	21–30
10	35%	4	73–81	1–4	1–3	1–3	4–11	39–47
10	40%	4	87–92	1–2	1–2	1–2	6–9	48–53
10	45%	4	126–143	3–15	3–12	3–13	22–39	82–99
10	50%	4	154–164	4–11	3–8	3–15 (3–17)	34–44	105–115
10	60%	4	241–261	>20000	88–101	88–101	91–111	182–202
10	70%	4	369–379	>20000	189–199	189–199	189–199	300–310
15	30%	4	145–170	1–2	1–2	1–2	5–18	78–103
15	35%	4	192–216	1–4	1–4	1–4	12–36	114–138
15	40%	4	277–312	14–38	10–15	11–19 (14–23)	56–87	188–223
15	45%	4	369–381	49–5029	18–104	40–105 (41–105)	111–125	269–281
15	50%	4	485–520	>20000	176–192	176–192	194–230	374–409
15	60%	4	780–805	>20000	416–437	416–437	420–445	646–671
15	70%	4	1216–1260	>20000	787–831	787–831	787–831	1059–1103
20	35%	3	452–466	2–9	2–8	2–9	102–111	313–327
20	40%	3	612–684	8–7132, >20000	4–218	4–218	194–266	453–525
20	45%	3	837–868	>20000	329–360	329–360	358–388	658–689
20	50%	3	1120–1140	>20000	533–585	533–585	580–600	921–941

Table 6: QWH instances, ranges over the instances of a cell. $|V|$: vertices of the QCP graph; completions: listed by exact cover, >20000 where the listing was capped (the span then also covers about 2000 sampled completions); span: dimension spanned by the completion indicators; rank of the Lovász optimum X^* found by CSDP, counting eigenvalues above 10^{-4} of the largest (in parentheses above 10^{-6}, where that differs); dim L: top multiplicity of the standard wrapper; $|V| - h + 1$: top multiplicity of the colouring wrapper of the cells.

ϑ was computed with CSDP from (3), one constraint per contradiction. A value below m is certified by $\lambda_{\max}$ of the wrapper recovered from the dual solution, which bounds ϑ from above by (2) whatever the solver's accuracy. The rank of a Lovász optimum counts its eigenvalues above 10^{-5} of the largest, or those before a drop by more than three orders of magnitude where there is one (only $179 \cdot 181$ has one above 10^{-5}), the multiplicity of $\lambda_{\max}(H)$ counts the eigenvalues within $10^{-5}\lambda_{\max}(H)$ of it, and the solutions were listed by exact cover search. The script is `bench/sat01.py` of the QUALEX-MS repository (commits b51234b and 33020d1), its output is in `bench/runs/sat01/`, and the largest problems, with 55000 constraints, took 27 minutes.

Table 7 answers the question of [Bu07] on these instances. After full propagation ϑ refutes four of the six primes that propagation leaves open, each certified by a wrapper with $\lambda_{\max} < m$, but the margin $m - \vartheta$ falls from 0.064 (14887) to $7 \cdot 10^{-5}$ (32687). On 32713 and 65521 $\vartheta = m$: by Theorem 6.12 they have vector solutions, of dimension about 160 and 200, and they are unsatisfiable instances with $\alpha < \vartheta = m$. On the graphs of light propagation $\vartheta = m$ for all ten primes from 14887 up, while it still refutes 101, so each of the four refutations needs the contradictions that contradiction analysis derives. Conversely, propagation alone refutes 32749, 32707, 32693 and 32653, whose light graphs have $\vartheta = m$. Neither method contains the other, and propagation followed by ϑ refutes more than either. The polyhedral shadow of Theorem 6.12, a fractional exact cover with $p_i + p_j \le 1$ on the contradictions, was feasible on all seven instances tried, among them the three with $\vartheta < m$ (101 light, 14887 full, Petersen light), so these refutations are semidefinite and not polyhedral. For perspective, the solver needs a

		light propagation			full propagation					
instance	guesses	n	m	$m-\vartheta$	n	m	contr.	blind	$m-\vartheta$	rank X^*
101	0	145	106	0.103*	refuted by propagation					
1009	0	316	217	0	refuted by propagation					
14887	1	716	478	0	510	365	22926	0.93	0.0638*	94
32717	1	868	575	0	568	404	29157	0.94	0.0345*	108
32719	1	866	575	0	615	425	30173	0.93	$8.9 \cdot 10^{-4}$ *	101
32687	1	866	575	0	638	435	34862	0.94	$7.0 \cdot 10^{-5}$ *	106
32713	1	871	575	0	612	419	29772	0.93	0	161
65521	1	955	631	0	713	482	41260	0.94	0	198
32749, 32707, 32693, 32653	0	865–872	575	0	refuted by propagation					
Petersen	0	63	18	0.235*	refuted by propagation					
J_5	2				265	38	14796	0.74	0	211
$GP(11,2)$	5				329	42	19316	0.73	0	269
Coxeter	83				585	54	50832	0.74	0	509
J_7	30				543	54	55123	0.79	0	465

Table 7: Unsatisfiable SAT01 instances: primes (factoring) and graphs without a Hamilton cycle. guesses: made by the SAT01 solver, 0 when propagation refutes the instance. After light and after full propagation: n variables, m equations, the contradictions, the share of them that share no equation (blind to H_A), $m-\vartheta$, and the rank of the Lovász optimum X^*. A 0 means $|m-\vartheta| \le 5 \cdot 10^{-6}$ with the dual wrapper's $\lambda_{\max}$ above m; *: certified by a dual wrapper with $\lambda_{\max} < m$.

single guess on each surviving prime: the instances are easy, and ϑ helps at the edge of what propagation does. The 17-bit primes keep 60000 to 63000 contradictions after propagation, beyond what CSDP solves on the machine used.

On the Hamilton cycle instances ϑ refutes only the Petersen graph ($17.765 < 18$ on its light graph), which propagation refutes anyway. The four non-Hamiltonian graphs that survive full propagation all have $\vartheta = m$: they have vector Hamilton cycles, of dimension 211 to 509.

On the satisfiable instances (Table 8) the solutions are the vector solutions of dimension one, and the question is how many others the Lovász optimum carries. The four 14-bit products of two primes have a unique solution and a Lovász optimum of rank one: the solution is their only vector solution, and solving the semidefinite program factors them. The optimum of $179 \cdot 181$ is still spanned by its two solutions, but those of $241 \cdot 251$ lie in an optimal face of dimension 214, of which CSDP's optimum has 205 eigenvalues above 10^{-5} (Table 9). On the cubic graphs the optimum spans exactly the 12 solutions of the smallest one (6 cycles, each in two directions) and far more than the solutions elsewhere: 152 dimensions against 30, 335 against 36, 213 against 60 on the dodecahedron and 409 against 26 on $GP(13,2)$, as on the QWH instances in the middle of the transition (Section 7.6). The optimal wrapper's top multiplicity equals the rank on all the graphs and exceeds it on three factoring instances (2 against 1, 18 against 2, 211 against 205), where complementarity is not strict at the solver's accuracy; the exact faces below settle the last two. H_A, which is Lovász-optimal on all of them by Proposition 6.11(c), is far more degenerate on factoring, with $\dim L_A$ between 145 and 236 against ranks between 1 and 205, but exceeds the optimum by only 8 to 24 dimensions on the Hamilton cycle instances.

Table 9 gives the smallest jams of Section 6.8. Only 36 to 38% of the contradiction constraints of the factoring instances are independent, against 91 to 99% on the Hamilton cycle instances

instance	n	m	solutions	span	rank X^*	mult. $\lambda_{\max}(H)$	$\dim L_A$
$143 = 11 \cdot 13$ (light)	180	132	2	2	2	2	49
$14471 = 29 \cdot 499$	511	367	1	1	1	1	149
$14631 = 3 \cdot 4877$	514	370	1	1	1	2	150
$15591 = 3 \cdot 5197$	501	359	1	1	1	1	145
$16041 = 3 \cdot 5347$	506	362	1	1	1	1	146
$32399 = 179 \cdot 181$	636	441	2	2	2	18	198
$60491 = 241 \cdot 251$	740	514	2	2	205	211	236
cubic, 16 vertices	57	30	12	12	12	12	29
cubic, 20 vertices	195	38	30	30	152	152	160
cubic, 24 vertices	403	46	48	36	335	335	359
dodecahedron	263	38	60	60	213	213	227
$GP(13, 2)$	481	50	26	26	409	409	433

Table 8: Satisfiable SAT01 instances after full propagation (143 after light propagation; full propagation leaves one equation in two variables). solutions: listed by exact cover search, the directed Hamilton cycles from the fixed vertex for the graphs; span: the dimension of their indicators; rank of the Lovász optimum X^*; multiplicity of the top eigenvalue of the optimal wrapper; $\dim L_A$: that of H_A. Every solution is anchored at $C = 0$ in the optimal wrapper to a relative $4 \cdot 10^{-5}$. For $179 \cdot 181$ the eigenvalues of X^* fall by a factor 5000 after the second and then decay smoothly from $5 \cdot 10^{-5}$ of the largest, the $\sqrt{\mu}$ tail of a solution that is not strictly complementary; counted above 10^{-5} they would give rank 5.

with 20 vertices or more, so the reduction pays on the former: $241 \cdot 251$ took 3 minutes against CSDP's 21, and 1000001, whose 124173 contradictions are beyond CSDP on the machine used, took 46. Of these, the elimination took 56 seconds, the choice of Γ' 4 minutes, and the rest went to 17 interior point iterations with a Schur matrix of order 45984 (17 GB). On the Hamilton cycle instances CSDP was up to twice as fast. For 14471 and $179 \cdot 181$ the same pair was solved over a basis of V from a dense eigendecomposition (`minjam.py dense`). The eigenvalues of Y fall by factors between 10^2 ($GP(13, 2)$) and 10^{10} (16 vertices) at the edge of R^*, the solutions lie in R^* to between $4 \cdot 10^{-13}$ and $2 \cdot 10^{-4}$ ($GP(13, 2)$ again), and BYB^T vanishes on every contradiction, imposed or implied, to $3 \cdot 10^{-9}$. On $GP(13, 2)$ a failing factorization ended the method early, and two directions at the edge, where Y has $2 \cdot 10^{-6}$ and W $4 \cdot 10^{-3}$ of their largest eigenvalues, were assigned to W by pairing the eigenvectors of the two.

The ranks agree with CSDP's on all the graphs, and R^* settles the two factoring instances on which CSDP left complementarity in doubt. On $179 \cdot 181$ the reduced problem is strictly complementary at dimension 2, so the multiplicity 18 of Table 8 counts a continuum of near-optimal eigenvalues of CSDP's wrapper. On $241 \cdot 251$, R^* has dimension 214, and its last directions carry eigenvalues near 10^{-6} of the largest at a point of maximal rank; CSDP's rank 205 and multiplicity 211 both cut through that edge.

The set Γ' that the elimination returns is independent but can be nearly dependent in floating point. On $241 \cdot 251$ an unpivoted Cholesky factorization of its Gram matrix meets squared pivots down to $4 \cdot 10^{-9}$ of the largest diagonal entry (`minjam.py cond`), and the interior point method broke down at $\langle Z, W \rangle \approx 10^{-6}$, when its Schur matrix lost positive definiteness. A pivoted Cholesky factorization of the Gram matrix of Γ' and up to 6000 further pairs (`minjam.py select`), with entries $(\Pi_{ik}\Pi_{jl} + \Pi_{il}\Pi_{jk})/2$ where $\Pi = BB^T$, picks a set of the same exact rank

instance	n	contr.	$\dim L_A$	$\dim V$	$\lvert\Gamma'\rvert$	$\dim R^*$	$\operatorname{rank} W$	time	rank X^*	CSDP
$14471 = 29 \cdot 499$	511	23319	149	2849	8326	1	148	1.5 min	1	5 min
$32399 = 179 \cdot 181$	636	34709	198	6605	13096	2	196	11 min	2	11 min
$60491 = 241 \cdot 251$	740	44477	236	11316	16650	214	22	3 min	205	21 min
$1000001 = 101 \cdot 9901$	1383	124173	472	65645	45983	445	27	46 min		—
cubic, 16 vertices	57	1010	29	12	423	12	17	1 s	12	1 s
cubic, 20 vertices	195	9025	160	4622	8258	152	8	24 s	152	25 s
dodecahedron	263	12673	227	13538	12340	213	14	2 min	213	54 s
cubic, 24 vertices	403	33335	359	31900	32720	335	24	18 min	335	8 min
$GP(13, 2)$	481	37848	433	56790	37171	409	24	21 min	409	12 min
cubic, 30 vertices	637	83128	581	87002	82069	not completed				
cubic, 40 vertices	1255	207199	1179	490160	205450	beyond the memory				

Table 9: The smallest jams R^* of Section 6.8, after full propagation: variables, contradictions, $\dim L_A$, $\dim V$ and the number $\lvert\Gamma'\rvert$ of independent contradiction constraints (from a dense eigendecomposition for the first two instances, by exact elimination modulo $2^{61} - 1$ for the others), $\dim R^*$ and the rank of the certificate W, which add up to $\dim L_A$, the time of the computation (on a machine shared with other runs, except for 14471, $241 \cdot 251$ and the cubic graph with 20 vertices), and for comparison the rank of CSDP's optimum (Table 8) and CSDP's time.

whose last squared pivot is $2{\cdot}10^{-5}$ of the largest diagonal entry on $241{\cdot}251$ and $4{\cdot}10^{-6}$ on 1000001. On the Hamilton cycle instances the set from the elimination is well conditioned as it is: no squared pivot falls below 10^{-2} on $GP(13, 2)$ or below $6{\cdot}10^{-3}$ on the cubic graph with 30 vertices. The cubic graphs with 30 and 40 vertices keep 82069 and 205450 independent constraints. The first fits in rectangular full packed storage [GWDL10], in 27 GB at about 8 minutes per iteration, but its run was not completed; the second does not fit. The computations are `bench/minjam.py` of the QUALEX-MS repository, with the exact elimination in `bench/srank.cc`; the logs of the runs are in `bench/runs/sat01/minjam/`, and each subspace R^*, with the optimum BYB^T and the certificate BWB^T, is in `<inst>.jam.npz` beside its instance in the ignored work directories.

8 Conclusions and open questions

For one clique, anchoring is always possible and puts the clique on the secular curve; its effect is local, with ascent towards heavier cliques in proportion to the gain per exchanged weight. Used on the solver's own best clique it is a reliable improvement step, used on a poor clique it is harmful. For several cliques, the free entries are far fewer in effect than in number: the levels are forced by the level identity, cliques of equal weight collide into the degenerate case, and the remaining system may force non-edges to be invisible or the level to vanish. Anchoring all maximum cliques at the top of the spectrum is possible exactly when $\vartheta = \omega$, and then it is what every Lovász-optimal wrapper does; this explains the degeneracy of the Lovász-optimal wrappers observed in QUALEX-MS, even though by Theorem 6.1 they are the best wrappers for the relaxed program. On odd holes and odd antiholes no nontrivial wrapper anchors all maximum cliques, and this characterizes perfect graphs through the strong perfect graph theorem.

The quasigroup completion graphs of Busygin and Pasechnik [BP03] show why the hard case cannot be put to work in general. For a completable instance the standard wrapper that

QUALEX-MS builds is itself Lovász-optimal, its degenerate top eigenspace contains all the completions, and they are exactly the nonnegative points of its sphere of global minimizers (Proposition 6.10); finding one of them is QCP. On random instances that eigenspace is often much larger than the span of the completions, and even the least degenerate Lovász-optimal wrapper exceeds that span in the middle of the transition from one completion to many (Section 7.6).

SAT01 [Bu07] carries this over to every problem that reduces to it (Section 6.5). The equation wrapper H_A is Lovász-optimal exactly when $\vartheta = m$; then its degenerate top eigenspace holds every vector solution, and the nonnegative points of its sphere of global minimizers are the exact covers of the equations, blind to the other contradictions. $\vartheta < m$ refutes an instance exactly when it has no vector solution (Theorem 6.12). On factoring and Hamilton cycle instances vector solutions appear early (Section 7.7). ϑ refutes small primes only on the graphs that SAT01 propagation has enriched with derived contradictions, with a margin that shrinks fast, and no non-Hamiltonian graph that survives propagation; propagation in turn refutes primes on whose unpropagated graphs $\vartheta = m$.

The NP-completeness is therefore not hidden in the wrappers. For a proper Lovász-optimal wrapper the maximum cliques are exactly the nonnegative points of the sphere of global minimizers (Proposition 6.13), and exact cover has such a wrapper, H_A, on every satisfiable instance, without any semidefinite program. Deciding whether that sphere has a nonnegative point is the exact cover problem itself (Corollary 6.14). The semidefinite program and linear algebra produce the polytope of nonnegative points of the degenerate eigenspace, and what remains NP-hard is finding a vertex of largest norm, which is the step from a Lovász optimum to a rank-one one. Contradictions outside the equations add improperness (Proposition 6.15), but that is not what keeps the problem hard. Nor do the quadratic conditions. The surface of the standard wrapper, which every solution lies on, cuts the sphere down to the solutions together with nonnegativity (Proposition 6.16), and its intersection with the sphere is explicit (Proposition 6.18). So everything but nonnegativity costs linear algebra and one scalar equation. In QUALEX-MS the surface helps the orthant (Section 6.7), but without nonnegativity it finds nothing, and the points of the intersection turn negative as soon as they leave a solution.

Nor does shrinking the degenerate eigenspace. The top eigenspace of every Lovász-optimal wrapper contains the subspace R^* spanned by all Lovász optima, and when $\vartheta = m$ the optimality of H_A turns the computation of R^* into a spectrahedron over L_A without an objective (Proposition 6.19). Exact sparse elimination and a small interior point method give R^* with a certificate of minimality, also on an instance beyond CSDP. R^* is the span of the solutions only on instances that are easy anyway; on the others it keeps 91 to 95% of L_A, and the Douglas–Rachford search gains nothing consistent from the smaller sphere (Section 6.8).

Questions left open:

1. Characterize the graphs whose maximum cliques can all be anchored at a nonzero level; the condition is not hereditary, and Theorem 5.4 settles only pairs.

2. Given that $\vartheta > \omega$, how few eigenvalues can lie above the common level of an anchoring? This is an eigenvalue optimization over the affine family of anchoring wrappers.

3. Is there a weighted analogue of Theorem 6.7, i.e. which weighted graphs have, hereditarily, anchorings of all maximum weight cliques?

4. What structure makes the Lovász optimum of `gen200_p0.9_44` full support and rank 157, and is it typical of generated instances with a planted clique?

5. On QCP graphs, when does the Lovász optimum span more than the completions? On the small instances of Section 7.6 it did only between the regimes of few and of very many completions, and none of them was hard for search. (An earlier version of these notes asked whether the hard case could be exploited by a search inside the degenerate eigenspace; Sections 6.4 and 6.6 answer that it cannot in general, unless P = NP.)

6. Which primes keep a vector solution after SAT01 propagation (32713 and 65521 did, while 32687 lost it by a margin of $7 \cdot 10^{-5}$), and does every prime from some size on keep one?

7. Contradiction analysis and ϑ are incomparable. Declaring x_j and x_k contradictory when the instance with $x_j = x_k = 1$, after propagation, has ϑ below its number of equations is a polynomial rule that subsumes contradiction analysis. How much does it add on the instances of Section 7.7, and does iterating it refute 32713?

8. The non-Hamiltonian cubic graphs tried all have vector Hamilton cycles, of dimension in the hundreds. Do they have an explicit description through the symmetries of the encoding, and which cubic graphs besides the Petersen graph have none?

9. On every instance of Table 9 the reduced problem of Section 6.8 is strictly complementary, $\operatorname{rank} Y + \operatorname{rank} W = \dim L_A$. Is it always? And is R^* always the top eigenspace of some Lovász-optimal wrapper, or can the second-order coupling in Proposition 6.19(d) force every optimal wrapper's jam to be larger?

References

[Ba95] A.I. Barvinok, Problems of distance geometry and convex properties of quadratic maps, *Discrete & Computational Geometry* 13 (1995) 189–202.

[BM03] S. Burer and R.D.C. Monteiro, A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization, *Mathematical Programming* 95 (2003) 329–357.

[Bo99] B. Borchers, CSDP, a C library for semidefinite programming, *Optimization Methods and Software* 11 (1999) 613–623.

[AGKS00] D. Achlioptas, C. Gomes, H. Kautz and B. Selman, Generating satisfiable problem instances, in: *Proceedings of the 17th National Conference on Artificial Intelligence (AAAI-00)*, 2000, 256–261.

[BP03] S. Busygin and D.V. Pasechnik, On $\overline{\chi}(G) - \alpha(G) > 0$ gap recognition and $\alpha(G)$-upper bounds, *Electronic Colloquium on Computational Complexity*, Report TR03-052, 2003 (`qcp.tex`).

[Bu06] S. Busygin, A new trust region technique for the maximum weight clique problem, *Discrete Applied Mathematics* 154(15) (2006) 2080–2096.

[Bu07] S. Busygin, The SAT01 framework for NP problems, note, 2003–2007 (`sat01theta.tex` in the SAT01 repository).

[BW81] J.M. Borwein and H. Wolkowicz, Facial reduction for a cone-convex programming problem, *Journal of the Australian Mathematical Society, Series A* 30 (1981) 369–380.

[CW] S. Busygin, A least squares framework for the maximum weight clique problem, note, 2007, updated 2010 (`CliqueWrapper.tex`).

[Co84] C. Colbourn, The complexity of completing partial latin squares, *Discrete Applied Mathematics* 8 (1984) 25–30.

[CRST06] M. Chudnovsky, N. Robertson, P. Seymour and R. Thomas, The strong perfect graph theorem, *Annals of Mathematics* 164 (2006) 51–229.

[DLLP08] L. Dupont, D. Lazard, S. Lazard and S. Petitjean, Near-optimal parameterization of the intersection of quadrics: I. The generic algorithm; II. A classification of pencils; III. Parameterizing singular intersections, *Journal of Symbolic Computation* 43 (2008) 168–191, 192–215 and 216–232.

[Ga57] D. Gale, A theorem on flows in networks, *Pacific Journal of Mathematics* 7 (1957) 1073–1082.

[GLS88] M. Grötschel, L. Lovász and A. Schrijver, *Geometric Algorithms and Combinatorial Optimization*, Springer, 1988.

[GWDL10] F.G. Gustavson, J. Waśniewski, J.J. Dongarra and J. Langou, Rectangular full packed format for Cholesky's algorithm: factorization, solution, and inversion, *ACM Transactions on Mathematical Software* 37(2) (2010), Article 18.

[HKR02] M. Halická, E. de Klerk and C. Roos, On the convergence of the central path in semidefinite optimization, *SIAM Journal on Optimization* 12 (2002) 1090–1099.

[Ho70] A.J. Hoffman, On eigenvalues and colorings of graphs, in: B. Harris, ed., *Graph Theory and its Applications*, Academic Press, 1970, 79–91.

[HRVW96] C. Helmberg, F. Rendl, R.J. Vanderbei and H. Wolkowicz, An interior-point method for semidefinite programming, *SIAM Journal on Optimization* 6 (1996) 342–361.

[JM96] M.T. Jacobson and P. Matthews, Generating uniformly distributed random latin squares, *Journal of Combinatorial Designs* 4 (1996) 405–437.

[Ka72] R.M. Karp, Reducibility among combinatorial problems, in: R.E. Miller and J.W. Thatcher, eds., *Complexity of Computer Computations*, Plenum Press, 1972, 85–103.

[Kn94] D.E. Knuth, The sandwich theorem, *Electronic Journal of Combinatorics* 1 (1994) #A1.

[Le76] J. Levin, A parametric algorithm for drawing pictures of solid objects composed of quadric surfaces, *Communications of the ACM* 19 (1976) 555–563.

[Lo79] L. Lovász, On the Shannon capacity of a graph, *IEEE Transactions on Information Theory* 25 (1979) 1–7.

[LPP06] S. Lazard, L.M. Peñaranda and S. Petitjean, Intersecting quadrics: an efficient and exact implementation, *Computational Geometry: Theory and Applications* 35 (2006) 74–99 (QI, `https://gamble.loria.fr/qi/`).

[Me92] S. Mehrotra, On the implementation of a primal-dual interior point method, *SIAM Journal on Optimization* 2 (1992) 575–601.

[MS83] J.J. Moré and D.C. Sorensen, Computing a trust region step, *SIAM Journal on Scientific and Statistical Computing* 4 (1983) 553–572.

[NÖ03] S. Niskanen and P.R.J. Östergård, *Cliquer User's Guide, Version 1.0*, Technical Report T48, Communications Laboratory, Helsinki University of Technology, 2003.

[Ov92] M.L. Overton, Large-scale optimization of eigenvalues, *SIAM Journal on Optimization* 2 (1992) 88–120.

[Pa98] G. Pataki, On the rank of extreme matrices in semidefinite programs and the multiplicity of optimal eigenvalues, *Mathematics of Operations Research* 23 (1998) 339–358.